

ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

REALTIME ETHIOPIAN SIGN LANGUAGE ALPHABET RECOGNITION VIA COMPUTER VISION AND MOTION SENSOR

A MASTER'S THESIS

By

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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING

OCTOBER 2021



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EYOB BOKRU BERHE

A Thesis Submitted as a Partial Fulfilment to the Requirements for the

Degree of Master of Science in

Electrical and Computer Engineering (Computer Engineering)

to

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING

OCTOBER 2021

Approval Page

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Declaration

I hereby declare that this thesis entitled "Real-time Ethiopian Sign Language alphabet recognition via computer vision and motion sensor" was prepared by me, with the guidance of my advisors. The work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted. In whole or in part, for any other degree or professional qualification.

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ii

Abstract

Unlike ASL or other Sign language alphabets, ESL fingerspelling is primely distinguished by hand shape and hand motion. ESL finger spelling represents consonant series with hand configurations, seven movements correspond to the seven vowels. This makes Ethiopian sign language dynamic. In dynamic sign languages, motion detection is essential for the recognition of the language. In this work, an ESL fingerspelling recognition using a fusing of computer vision and motion sensor research work is proposed to recognize and classify the vowel using motion sensor data and deep learning, also using computer vision to recognize consonants. With A large dataset of Ethiopia manual sign language alphabets from multiple subjects, two deep neural networks selected to derive a best topology for each. The vison model is implemented by a YoloV5 model and the smartwatch motion sensor motion model is implemented by LSTM network. Several experimental tests were conducted to gain the most accurate result. And the proposed model yolov5 and Bi-LSTM got 98.5% and 97% respectively with inference time less than 2 Msec. As a result, it has been proven that the proposed decoupling model would be effective for Realtime EMA classification.

Key word: EMA, YOLOv5, fusion, smartwatch

Acknowledgments

First, I would like to thank my Almighty GOD for everything during this hard time, and my thesis advisor Dr. Solomon Zemene for his support and guidance through the process of completing the best of this research. I would also like thanks the experts and students who were involved in the data collection process for this research work. Last but not least, I would like to thanks Addis Ababa Science and Technology university, my family and friends who gave me persistent encouragement and support while I was working on my thesis.

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List of Abbreviation and acronym's

ANN	Artificial neural network
ASL	America sign language
CNN	Conventional neural network
CSPNET	Cross Stage Partial Network
EthSL	Ethiopia sign language
EMA	Ethiopia Manual alphabet
ENAD	Ethiopia National Association of the deaf
IMU	Inertial Measurement Unit
LSTM	long short time memory
MLP	Multilayer perceptron
MOE	Ministry of Education
PANet	Path Aggregation Network
ReLU	Rectified linear unit
RNN	Recurrent neural network
SVM	Support vector machine
UUID	Universally unique identifier
WHO	World health organization.
YOLO	You only look once

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CHAPTER 1 : Introduction

1.1 Overview

According to Britannica, language is A System of conventional speaking, manual (signed), or written symbols using which human beings, as members of social groups and participants in its culture, express themselves. One form of language is sign language expressed in manual form. That is used commonly among hearing-impaired communities[1]. The hearing-impaired community includes hearing-impaired people, families of the hearing-impaired person, and other people who can communicate with the hearing impaired.

Ethiopia has around 1.5 mil people with deafness, according to the Ethiopian National Association of the Deaf report in 2007 [2]. This number is approximate and expected to rise in recent years. According to WHO, around 466 million people worldwide have hearing impaired, and among those 34 million of these are children [1], to support the above argument, according to speech from miss Hana Bekele, general manager of Ethiopia National Association of the deaf (ENDA) the number of hearing-impaired people in Ethiopia is expected to be around 5 million. In addition to the Ministry of Education (MOE), only 4 % of Ethiopian children with disabilities. Attend primary schools due to stigma among parents and education inaccessibility, rigid teaching practices, poorly trained teachers, and lack of adapted learning resources. The Study of (EthSL) presents several and advantages. It aids the development and recognition of the language itself, reinforces the fact that it is a real language, and influences the deaf community by enhancing educational and employment opportunities for its members.

In [3] states that sign language is the language that is spoken among the deaf community. That uses the hand, which includes fingers, hand and finger, and the body, head, and face. Recognition and understanding of sign languages is thus an entirely visual and temporal process performed by a human being. As mentioned in the report above, there is the

difficulty of communication with those. We cannot understand sign language. This presents a severe language barrier is presented.

Thirty-four hand shapes represent the consonants in the Ethiopian Manual Alphabet (EMA) which is modeled on Fidel, an Ethiopic script, and there are several forms of vowels, each of which distinguished by moving the consonants different, almost all hearing-impaired students learn both EMA and ASL fingerspelling at school in Ethiopia. These alphabets are used to spell word names and meanings which do not have single sign words. [4]Before 2009 we used to have 33 Fidel's. Since then, we have added one additional letter (ⷘ) that was introduced. Sign language s convey meaning in many forms simultaneously, which poses the problem that they use multi-channel.

Sign language gestures can be classified into three categories: a static gesture is a particular handshape, orientation, and pose. That is used to fingerspelled gestures of alphabets and digits. The dynamic gesture can be expressed as a static gesture with movement this is used to represent isolated words. The last one is continuing gestures sequences of dynamic gestures used to represent sentences in sign language.

In EMA are dynamic gestures with 34 base letters where each letter (consonants) has six other variations representing the vowels [4]. The vowels are bound with timing, placement, and orientation. In general, the base letters are static, but to create the seven syllables, there is moving in hand. This makes the EMA a dynamic gesture.

In Figure 1.1. B. shows the written form that is used in Amharic/ geeze in general, and c. represents the corresponding sign movement corresponding to the seven vowels. Example to represent the second Fidel in He(υ) signer have to move his/her hand to left while maintaining the HE(υ) handshape in a.

Including hearing-impaired people in the common work has been very challenging in the world. Mostly as a result of a lack of communication. Nevertheless, the gap is partially solved by other researchers who have been looking into ways to create a new application that lifts the communication gap in different sing languages.

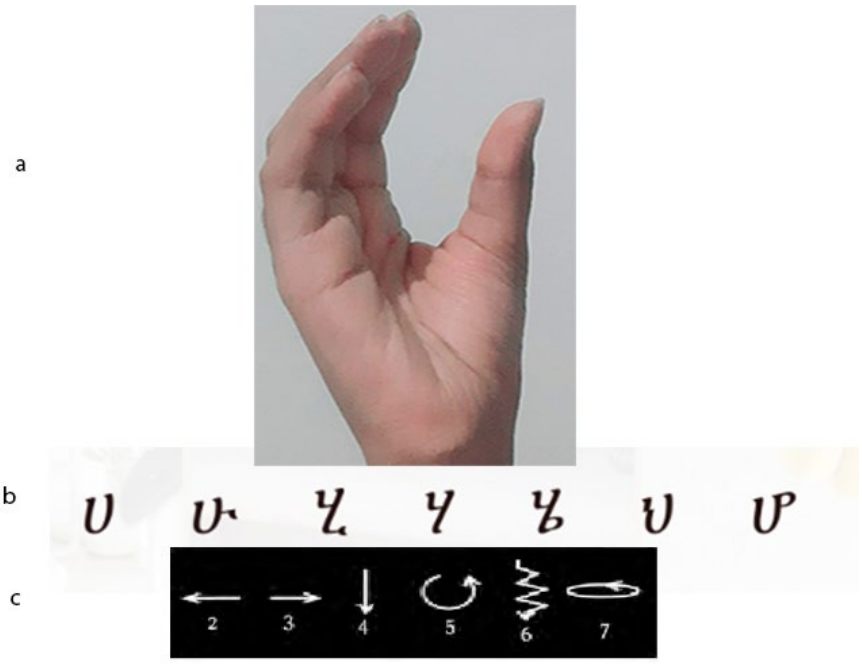


Figure 1. 1 An example of Ethiopia manual alphabet a. represents base alphabet i.e., ሀ

However, less development and research work have been done in ESL. Moreover, the field of study is booming, but it has not fully matured. We feel that working on this will help to further research and development.

1.2 Motivation

Disable persons are part of our community and family. All of us should participate in making their life exciting, fun, let them receive an equal opportunity with everyone. As a computer engineering researcher, we have related my field of study to this major concern of communication barriers. Today's technology advancements have many advantages for sign language and language in general. The area of artificial intelligence, especially computer vision, is one of the technologies along with machine learning, and sensors are emerging technologies that have been benefiting and will benefit Sign language detection and recognition. This thesis on realtime EMA recognition will play a role in advancing the study of Ethiopian sign language.

1.3 Problem Statement

The development of certain computer science and engineering disciplines, like deep learning, machine learning, pattern recognition, image processing, data sensing, IoT, etc., promises the required technical support to tackle various issues in computer vision and human-computer interaction[3], [5]–[8]Many technologies have been applied using a camera, Gloves, Kinect, leap motion, depth camera, arm sensors, armband, etc.

Various studies have been carried out to develop a system that converts sign language to text or vice versa for various SL all over the world. [10] Examples include American sign language, South African sign languages, Chine's sign language, Indian sign language, British sign language, and Bangladeshi sign language [10]. This shows there is potential to automate using techniques such as image, device, and machine learning statistical methods.

Dynamic sign language/gesture is a moving gesture, represented by sequence image or video processing consists of hand orientation and tracking this will be calculated every frame this requires high bandwidth and as the main properties of sign language that is movement emphasis, signer dependent variables, occlusion, and so on. This makes it inconvenient for realtime at this time. To address such issues, it is essential to solving this using a different technique that separates the base letter itself from it the movement that defines the vowels, the proposed technique for ESL alphabet recognition via computer vision and motion sensors. This proposal was implemented using the Ethiopian manual sign language alphabet that is dynamic gestures.

Research Question

The following research questions have to be answered.

- **RQ1** What is the current realtime Ethiopian sign language alphabet recognition systems status, and which algorithm is best suited for a realtime system?
- **RQ2** What are the state-of-art Vision and sensor-based systems? Will decoupling dynamic gestures gesture from its movement help to dynamic sign language recognition?

1.4 Objective

1.4.1 General Objectives

The general objective of this study work is to design and implement a recognition model EMA fingerspelling using object detection models, motion sensors, deep learning, and an LSTM network.

1.4.2 Specific Objectives

The specific objectives are:

- Collect the desired motion data by programing the smartwatch
- Collect images of EMA hand sign
- Prepare dataset for model training and model testing
- Develop recognition and classification model
- Evaluate the performance of the developed model using accuracy, precision, f1-score, recall, and detection rate

1.5 Research Methodology

To conduct this research work, the methodologies mentioned below will be used

- Literature Review: This is a comprehensive examination of previously published related works. Before beginning the actual work, a thorough review of the literature in this field was conducted in order to gain an understanding of the work. Many previous papers on sign language and gesture recognition were reviewed to better understand the most up-to-date techniques and methods in sign language and Ethiopian sign language. We used these reviewed papers in order to understand how other countries recognize and classify their sign language, such as Chinese, Indian and American. Furthermore, we try to concentrate on papers that deal with realtime classification and recognition. The tools and algorithms for developing the application were chosen based on information gathered from this paper.
- Data collection

The first task in this research was to figure out how signers spell each alphabet sign. Thus, help to collect data collection carefully. We have two types of data in this study

- Video data: captured using mobile phone from the trained participant. The content in those videos is handshape and orientation of each alphabet.
- Motion data: accelerometer and gyroscope data are the gesture of vowels collected using a smartwatch.

- Image preprocessing

  In this stage, the images selection, labeling, i.e., drawing bounding boxes in the images to specify an area of interest for the object detection models.

- Tools
  - Mobile camera
    - This was used to collect the handshape of the alphabet for a subject. Thus, participants were 40, and each subject was subjected to bring a 5-sec video recording
  - Smartwatch
  - Android studio
  - Google collab
  - LabelImage
  - TensorFlow Keras
- Experimentation and Testing

  To ensure that the developed system is effective, this will aid us in determining the system's strengths and weaknesses. Thus, future researchers may continue the work afterward.

1.6 Significance of the thesis

The research will aid in the effort to solve the problem of hearing-impaired people's communication with hearing people. The study has made a significant contribution to EMA's realtime recognition. The following are the advantages of this study. This research is significant in its theoretical and practical contribution to the existing body of research

knowledge. The theoretical contributions are the academic support of the theory of fusing image and motion data to decompile the isolated sign to make systems realtime.

1.7 Scope and Limitations of the thesis

The scope of this study is limited to the role of predictive analysis of Ethiopia sign language data's image and motion sensor using a state-of-the-art object detection deep learning model and LSTM. EMA data will be collected manually from a trained signer that can perform the sign of Ethiopian manual alphabets. For the quality of the data, we try to make it look like sign language data sets in Kaggle. The standard performance of measurement like accuracy, recall, precision, f1 score, MSE. Are considered in this research.

The major limitation of this research was the lack of budget. The data collected from the trained signers and the sign language community was a little difficult. The system will predict the sign with approximate hand shape and orientation. In the era of data mining, quality plays a great role in designing effective intelligent systems. Therefore, we have employed professional signers to train volunteer participants for a limited period to collect quality data, also, as the quality and size of data increase, the model's reliability, robustness, and performance increase. Thus, to collect more data with better quality, the research must employ professional signers for a prolonged period which is very difficult within the allocated time and budget. Also, the study is limited to Ethiopia manual alphabet only.

1.8 Contribution of the thesis

The specific contributions of this thesis include

- The development of a large Ethiopian Manual alphabet dataset in this study will be useful for future research in this area. The dataset used in this study was robust in terms of size, the number of participants, diverse background images.
- Another novel contribution of this study is the fused model of vision and motion data to recognize EMA in realtime. This is the first study to fuse Vision and device-based approaches for realtime EMA. Fingerspelling can be classified with high accuracy.

- Another contribution is recognition of complex backgrounds on videos and images of signs

1.9 Organization of the thesis

The rest of the thesis is structured as follows:

- Chapter 2 presents a detailed discussion on the meaning of the hearing-impaired, communication, sign language structure, fingerspelling, singing, Ethiopian fingerspelling, and Ethiopian sing languages. It also provides a review of related works in video and sensor-based sign language detection methods, in addition, desiccation on image representation, image and deep learning/ object detection techniques like YOLO and LSTM,
- Chapter 3 gives the proposed mechanism for dataset collection and classification algorithms, and efficiency improvement techniques.
- Chapter 4 will discuss the result obtained using the selected models' algorithms and smoothing implements results with a different experiment performed using various
- Chapter 5 concludes the work and presents recommendations for future improvements.

CHAPTER 2 : Literature review

2.1 Introduction

This chapter covers details about how the hearing-impaired people communicate and the sign linguistics, sign language architecture, specifically Ethiopian sign language architect, the grammar, discusses sign language recognition techniques and recommends the best practice for EthSL alphabet recognition.

2.2 Hearing-Impaired Community

Communication for deaf individuals can occur through visual or tactile modes (for individuals who are deaf and blind). Let's start by defining a hearing-impaired individual as someone who can't hear as well as someone that has hearing thresholds of 25 dB or higher in both ears. Individuals' hearing loss can be minor, moderate, severe, or profound. It might impact in both ears. As a result, it becomes hard to hear(listen) casual conversations or loud noises. [11]. Common visual communication modes include in sign language are gestures, cued speech, and lip-reading,

Ethiopia's hearing-impaired live in the same way as any other hearing-impaired person in their respective cultures, but they are unable to communicate with their community. The vast majority of hearing-impaired Ethiopians, who live in a rural part of the country, spend their lives in extreme separation. They are looked down upon as mentally deficient because of their lack of spoken communication. In many places, they are misunderstood as a result of their parents' sinful behavior or some form of a supernatural curse. in consequence, they are frequently isolated from the rest of the world, communicating only with their families or close friends via a limited amount of writing or signing, if they are able. More awareness of the hearing-impaired has been established in cities(deaf). Many parents want their children to attend school, but the resources available are insufficient to accommodate the number of potential pupils, missionaries, non-governmental organizations, and, more recently, schools. Several hearing-impaired schools have been established by the government.

2.3 Sign Language

Communication is the process of communicating information, thoughts, opinions, and expressions between individuals, both verbally and nonverbally. Hand gestures are nonverbal communications modalities that are used in conjunction with spoken communication. Hand gesture communication that is more ordered. [12]The use of one or both hands and facial expressions and body motions to create meaningful gestures is known as sign language. A one-handed sign language is made up of signs using the movement of one hand, whereas two-handed sign language is made up of movement of both hands. Facial expressions in signing contribute vital information to expressing to the moving part of the gesture. A word in spoken language is represented by the term sign in signing. Frequently, a sign represents a frequent word. The signer will be obliged to spell the term using fingerspelling if it is uncommon or not part of the collection of words commonly used in daily conversation. Sign language is a language that conveys meaning through physical communication and body movements. This can include incorporating hand forms, hand orientation and movement, body, arm, and facial gestures all at the same time to convey the speaker's opinions smoothly [13]. Because it is a visual language, speaking and hearing talents are not required to use it. Almost every country on the planet has its own sign language.

There is a difference between sign languages that are used within the same country, just like the difference of dialect in verbal languages. As to Wikipedia in the world, there are approximately 200 recognized sign languages [14]. Like in a spoken language, they have vocabularies, grammar, and spelling (fingerspelling) and have comparable language structure with spoken language. They share vocabularies, syntax, and spelling (fingerspelling) with spoken languages, and they have a similar linguistic structure. It is possible to express hearing-impaired people's culture, sign a song, convey poetry, and do whatever spoken language can do with sign language

2.4 Ethiopian Sign

Sign language dialect in Ethiopia corresponds to each spoken Amharic or Tigrigna word. The majority of Amharic/Tigrigna grammar is used to sign all parts of the language. When

employing signed Amharic/Tigrigna, people usually speak simultaneously to give the hearing-impaired individual the advantage of both signed and spoken word.

2.5 Ethiopian Sign Language Finger Spelling

The Ethiopian Finger spelling was developed by the ENAD in 1971, and the letter was approved by the ministry of education. It is now taught in all deaf schools alongside. American finger spelling. The finger spelling of the United States, when they want to convey words or thoughts in English or any other local language that is expressed though Latin script, such as afar, Oromifa, Sidam etc., they employ local finger spelling, when to communicate words or concepts in language that employs the geez alphabet, such as Amharic and Tigrigna [17], Ethiopia finger spelling is utilized. Ethiopia's manual spelling has 34 unique hand orientations and shapes to represent the first-order alphabets. By adding six different movements on the first order alphabet, the rest six order alphabets for each first order alphabets order alphabets are constructed.

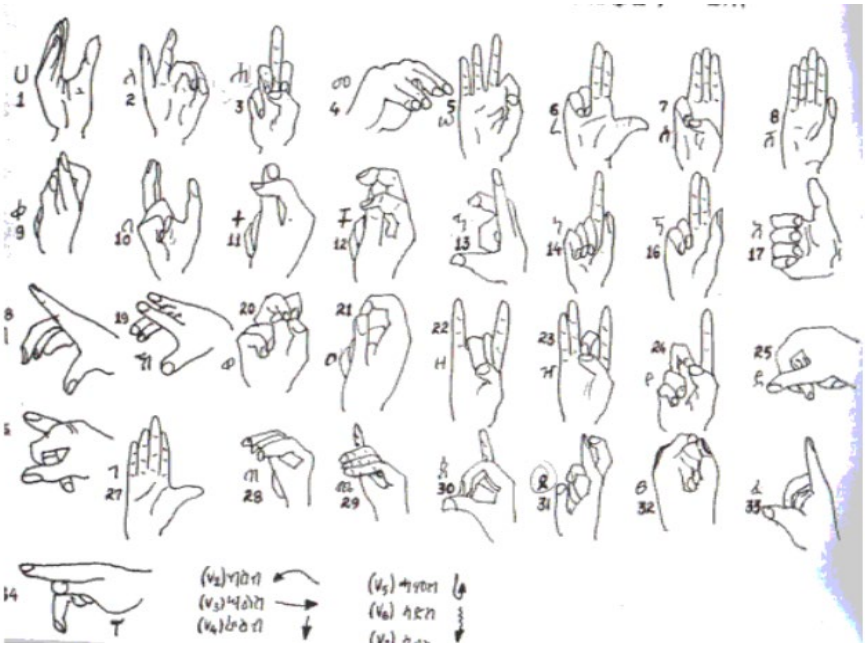


Figure 2. 1 Ethiopian manual signs finger spellings [1]

All hand shapes of EM alphabet this represent base shape of all alphabets

2.6 Sign Recognition Techniques

Automatic sign language recognition research is required to better communication between the deaf and the hearing. Sign recognition research is 30 years behind voice recognition research. This is because two-dimensional videos are far more complicated than one-dimensional voice signals. There is more explanation for why sign language is far behind, and sign language is not thoroughly researched [18]

Hand signs can be categorized into two main parts: static (non-tracking-based) and dynamic (tracking-based). Static gestures are those that do not use the movement of the hand to be recognized, and an open hand could be a simple example. in contrast, the dynamic gesture is those that include movement of the hand. E.g., moving the palm to the right or left or up or down.

Classification, sign language recognition systems are used to translate sign language into text or speech so that individuals who do not know these gestures can communicate with them. Typically, the systems are designed to recognize hand configuration, such as position, orientation, and movement. [18]. As a result, these configurations are recorded in order to deduce their related meanings. Based on the input capturing mechanism. Previously developed system used electromagnetic gloves, data gloves, leap motion, Microsoft Kinect, arm bands etc.

There are different forms of recognition approaches in the field of pattern recognition. Basically, they are divided into two sub-areas: Vision and device-based approaches. Vision-based techniques are those that use different types of cameras to obtain information about the hand. The techniques are those that use different types of cameras to obtain information about the hand. [6]Web-cameras and 3D depth cameras are examples of cameras used in this kind of approach. A lot of vision-based sign recognition techniques have been introduced in recent works; Sensor based techniques include all the methods that use gloves equipped with sensors to obtained information about the shape of the hand.

2.6.1 Device Based Recognition Approach

Device based approaches are data acquisition techniques that use data gloves equipped with sensors to map the human hand. In this section glove-based, handshape recognition techniques will be presented.

Glove-based handshape recognition involves typically the user wearing device which serves as a glove and a connecting cable that helps as an interface to a computer. This makes these methods a non-natural and challenging way to interact with computers [7]. Sign language information acquisition devices use electromagnetic-based interference to get interference to get information about the hand, which is sufficient to provide a description of a hand gesture. Data gloves are referred by researchers in different ways, e.g., Cyber-Glove and Accele Glove. But these techniques are mostly used for the recognition of alphabets in sign language.

2.6.2 Vision Based Approaches

In vision-based mechanisms, different kinds of normal cameras are ways to obtain information about the positions of hands and figures. This section presents different approaches that use various camera types like stereo cameras, webcams, and time of flight cameras to obtain information about the gesture. These methods utilize only a camera, which allows us the easiness and realization of free interaction between mechanisms without using extra costly devices.

This extra device free mechanism has its own problems like light sensitivity, different kinds of background environments, among the few makes the system very different to work in real-time environment. Other than this real-time scenario, these system accuracies and efficiency must be taken into consideration. This precondition makes the system very difficult to implement in real-world scenarios. Vision-based sign gesture capturing technique could be used. The input for these approaches is provided by RGB or RGBA images to be extracted for video cameras, time of flight-based cameras, and depth-sensing cameras like Asus Xtion and Microsoft Kinect. In Gesture-based human-computer

interaction (HCI) realtime video stream of the gesture and efficient and realtime responsive algorithms to capture and process gestures [10].

These techniques do not depend on a single camera, instead they depend on multiple cameras/sensors to obtain depth information about the sign being performed [19]. The use of 2d video-based algorithms in tracking the required depth information of hand trajectory leads to confusion when there is occlusion of hands.

2.7 Neural Networks

According to Rosenblatt, the perceptron is the "basic probability paradigm for information storage and organization of the brain." Modern machine learning applications rely on it. The perceptron is a simplified version of a hypothetical nervous system that emphasizes the essential qualities of an intelligent system that can learn without the uncertain situations that biological organisms frequently show. Early brain models that responded to stimuli sequences in order to accomplish particular algorithms using aspects of symbolic logic and switching theory were used to construct the perceptron [20].

In modern times, the perceptron is useful in classifying linearly-separable, binary data samples. A binary classifier maps real-valued vector input x to an output value f(x), which satisfies the following

$$f(x) = \{1, if\ wx + b > 0 \quad (2.1)$$

$$0, otherwise$$

In the Figure 2.2, each of which represented by an element of the vector x, and a bias input of 1. Each input connection has a weight parameter, which leads to the dot product of the combined input and weight, including term

Where w is a real-valued vector of synaptic weights, w, x is the dot product $\sum_{i=1}^{m} w_i x_i$, m is number of inputs, and b is the biased input. Thus, the $f(x)$ classifies the input as a positive or negative instance. Linearly-separable data can be separated into classes by at least a single 2- dimensional boundary line or a multi-dimension.

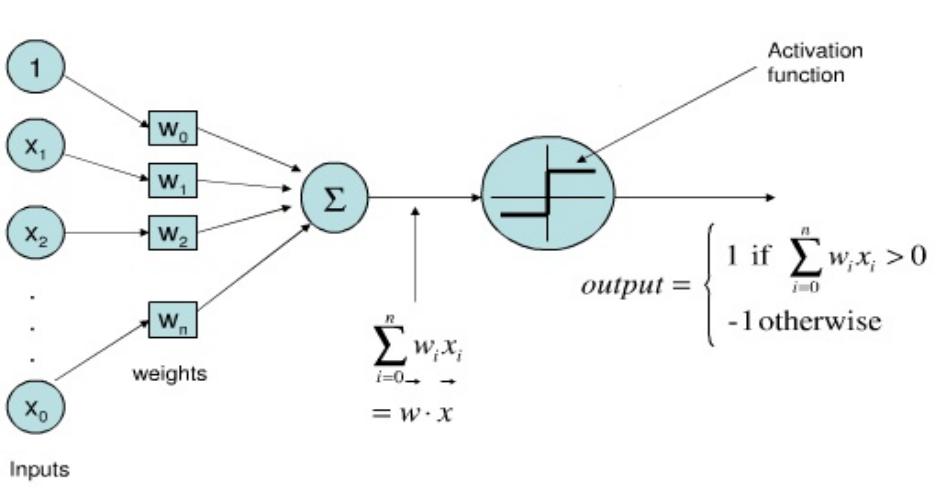


Figure 2. 2 A perceptron model with n real-valued inputs, source [20].

hyperplane. In binary classification, this means that only one decision boundary is necessary to properly separate the data samples in the sample space.

Perceptron was designed to mimic a single decision boundary; nevertheless, it fails to classify more than two linearly separable classes in multi-class classification. To tackle the challenge of multi-class classification, it was discovered that several perceptron layers could be chained together, and the outputs of the first perceptron layer could be fed forward as the inputs of the second perceptron layer.

A feedforward neural network is an artificial neural network (ANN) with only one direction of input flow and no cycles. The perceptron is a feedforward neural network in the sense that inputs are sent directly from the input layer to the output layers. The majority of feedforward neural networks are multilayer, fully linked neural networks. Each neuron in a layer is connected to every other neuron in the layer before and after it. The input is delivered on the first layer that is visible. The last visible layer produces the output. Hidden layers are the layers in between. Hidden layers allow the network to generate more abstract representation and avoid over-fitting.

In a neural network, there are a variety of neural activations that can be used. To simulate a goal response the changes non-linearly with explanatory input variables, the majority of these activation functions are non-linear. Non-linear combinations of functions provide output that cannot be replicated by a linear combination of inputs in this way. Some

noteworthy activations are Logistic (soft step), Rectified Linear Unit, Hyperbolic tangent(tanh) and SoftMax.

The backpropagation algorithms were created in order to train feedforward neural networks. The backpropagation algorithm is based on a cost or loss function that transfers values from one to more variables to a cost associated with a specific event, such a network prediction. the loss function should compute the difference between the expected output for an example and the result of propagating an input example through a neural network. The propagation and weight update must be stepped in backpropagation.

The propagation and update stages are performed until the network error can no longer be tolerated. The quality and speed of the learning method are affected by the fraction of the weight gradient eliminated. The fraction of weight gradient subtracted from a given weight is the learning rate. To compute the least objective function, we have optimization methods such as gradient descent, stochastic gradient descent, RMSProp, and AdaGrad.

2.7.1 Recurrent Neural Networks

The RNN is a directed cycle ANN with information flowing in multiple directions and at least one feed-back link between neurons. Because if produces an internal state of memory important for learning time-dependent sequences, this sort of network exhibits dynamic temporal behavior. A MLP is one of the simple types of recurrent neural networks, having outputs from one hidden layer coming back in to the hidden layer over a discreate time scale determined by delay unit.

Despite their utility, simple RNN units often use standard activation functions that struggle to capture long-term relationships due to vanishing gradients. With increasingly higher short-term dependencies, these gradients tend to obscure the importance of long-term dependencies. Gradient clipping and derivative methods are two significant ways that have been used to solve these gradient difficulties.

Simple RNN units' activation functions aren't as reliable for modeling long-term dependencies as other methods that follow critical events over time. By adding a memory cell to simple RNN units, the LSTM add intermediate storage. Using gated logic, the LSTM

memory cell was developed to preserve a state of extended dependence. The memory cell was constructed from smaller cells to create a composite unit with gate and multiplicative nodes. The LSTM's components are listed below[19].

- Input Node(gc) It uses the activation from the input layers x(t) at the current time step, as well as the activation from the hidden layer at the previous time step h(t-1) along recurrent edges. On the summed weight input, a squashing function is applied.
- Input Gate: A gate is a sigmoidal unit that receives activation from the current data point x(t) and the preceding time step's hidden layer activation. The gate values are multiplied by the value of another node.
- Internal State: Each memory cell has a sc node with linear activity, which the authors refer to as the internal state. A recurring, self-connected edge with a fixed weight of 1 exists in this condition. Because of the constant weight, the mistake doesn't explode or dissipate across numerous time steps. This includes fc's influence on the forget gat.
- Forget Gate: When necessary, these gates allow the network to flush the internal state contents. These are beneficial for networks that are constantly running. The influence of the foregate gate in the update of the internal gate state is shown in the update equation for the internal state in figure.
- Output Gate: The generate output value, vc, the output gate, oc, multiplies the internal state value, sc. The internal state value is sometimes processed through a non-linear activation function to give the output values the same range as standard hidden units with squashing functions, although depending on the circumstance, this may be skipped.

Because of their cyclic nature, RNNs are more difficult to train with feedforward neural networks. To deal with the network's complicated, cyclic structure, it must first be transformed into a deep feedforward neural network over time [21]. after unfolding, the training proceeds in a backpropagation, the training patterns are visited in a specific order, which is one of the differences

2.7.2 Deep Learning

Deep learning is a subclass of machine learning which is the learning process is based on an artificial neural network. And the learning paradigm can be supervised or unsupervised [22] Deep learning gives computers or machine ability to perform some tasks human being, or enables computers to perform like human beings by learning them, learn by example. Deep learning is key technology behind different AI application areas such as driverless cars, speech recognition, Natural language processing, and other different related tasks. Deep learning is a type of artificial intelligence that imitates the working principle of the human brain to process input data and making useful patterns for use in the decision or make a classification. In deep learning the model learns to perform classification directly from the input image, sound and text. And the model shows a remarkable result inaccuracy sometimes it achieves a result which exceeds the human level.

Representation learning is a set of methods that allows the machine to process in a raw form and discover useful patterns for detection and classification tasks. And deep learning is one type of representation learning with multiple level of representation [22]. Deep learning uses an artificial neural network. The term deep is used to refer to the number of hidden layers in neural network. When an artificial neural network contains many hidden layers, the network is called deep architecture. Term deep learning refers to the deep architecture of the artificial neural network [22] [23]. The standard machine learning algorithms require a separate feature extractor which is optimized by the programmer after that learns from the extracted feature.

Deep learning architecture such as deep belief networks, deep neural networks, convolutional neural networks, and recurrent neural networks have been used in different application areas to embrace natural language processing, speech recognition, computer vision, audio recognition, etc.

2.7.3 Object Detection Algorithms

Object detection is commonly used in reality. Target detection is to find all objects of interest in the image. It includes two subtasks: determining the category and location of the

object. We divide target detection algorithms into traditional methods [19]and methods based on convolutional neural networks.

Traditional object detection algorithm:

The traditional object detection algorithm mainly consists of three steps:

- Using sliding window frames of different sizes and proportions to slide on the input picture with a certain step length and as a candidate area;
- Perform feature extraction on the local information of each candidate region; traditional methods include color-based, texture-based, shape-based methods, and some middle-level or high-level semantic features. The final target that needs to be detected is the final output result of the algorithm.
- Use classifiers for recognition, such as SVM [10] models.

After the check box is judged, a series of candidate boxes that may be the detection target will be obtained, and these candidate boxes may have some overlapping conditions. At this time, NMS [25] is needed to merge the candidate frames. Although traditional detection algorithms can achieve good results in specific scenarios, in complex environments, such as weather changes, uneven distribution of workers, and different helmets, its accuracy is difficult to guarantee.

Its generalization ability is poor. Besides, traditional manual design features require a lot of prior knowledge. The three-part detection process is cumbersome and computationally expensive and cannot meet the realtime detection.

Object detection methods based on convolution neural networks are mainly divided into two categories: one is a two-stage approach represented by RCNN [25]The other is a regression-based target detection algorithm represented by YOLO[26]

In theory, the two-stage algorithm is more accurate than the one-stage algorithm. Represented by Faster-RCNN[27] , the two-step target detection algorithm consists of a convolutional layer, a region candidate network, a region of interest pooling layer, and a classification layer.

The one-stage target detection algorithm truly achieves end-to-end training. Represented by YOLO, the regression-based target detection algorithm completes the target category's determination and the positioning of the target at one time. The entire network structure is composed of only convolutional layers and the input image. After the convolution operation, the target category and position are directly returned. Therefore, the one-stage target detection algorithm is faster than the two-stage target detection algorithm, especially YOLOv5, which has reached an advanced speed and accuracy level.

the network structure of yoloV5 is divided into three parts, backbone, neck, and output. In the backbone, the input image with 640×640×3 resolution goes through the Focus structure. Using the slicing operation, it first becomes a 320×320×12 feature map, and then after a convolution operation of 32 convolution kernels, it finally becomes a 320×320×32 feature map. The CBL module is a basic convolution module.

A CBL module represents Conv2D + BatchNormal + LeakyRELU. The BottleneckCSP module mainly performs feature extraction on the feature map, extracting rich information from the image. Compared with other large-scale convolutional neural networks, the BottleneckCSP structure can reduce gradient information duplication in convolutional neural networks' optimization process.

YOLOv5 is small and fast, requiring fewer computing resources than other current state-of-the-art architecture models while maintaining accuracy close to that of current state-of-the-art detection models. It is far faster than other YOLO devices. To extract the feature map from the image, YOLOv5 employs CSPNET[28]as the backbone. It also makes use of the Path Aggregation Network (PANet) [29] to improve data flow. The architecture of YOLOv5 is depicted in the following image [29] .

There are three sections to it.: (1) Backbone: CSPDarknet, (2) Neck: PANet, and (3) Head: Yolo Layer. [26]The data is fed into CSPDarknet for feature extraction before being fed into PANet for feature fusion. Yolo Layer then outputs detection results (class, score, position, and size).

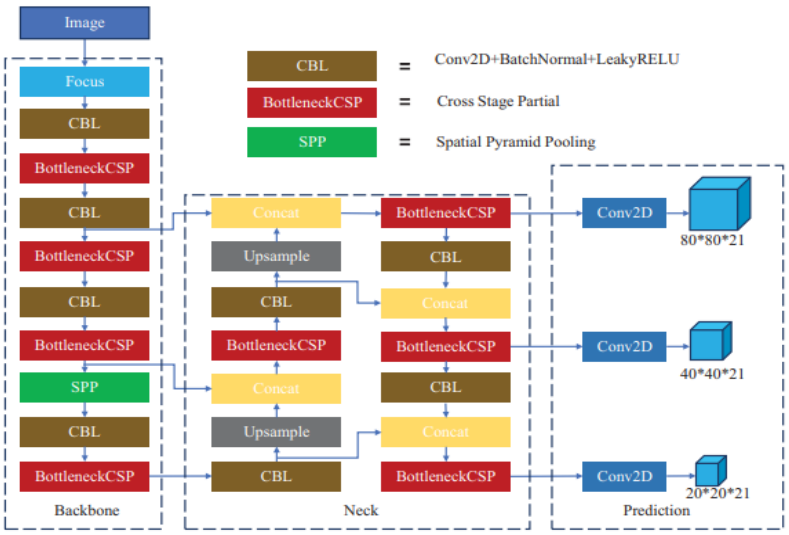


Figure 2.3 The network architecture of Yolov5.

YOLOv5 is used for the following reasons:

- Yolo is the fasters makes it convenient for Realtime
- Has useful components such as state-of-the-art activation function, hyperparameter, data augmentation technique and a convenient manual.
- Its lightweight architecture makes it computationally easy to train with small resources.
- The size of the model is quite small and lightweight, thus can be used with mobile devices.

2.8 Realtime Ethiopian Sign Language Alphabet Recognition

Ethiopia Manual alphabets are dynamic signs. While designing a solution, dynamic sign recognition systems must consider hand shape, hand orientation, hand location, and hand movement. The system analysis is mainly on hand shape and motion trajectory. In order to design a system that solves in realtime, we propose to decouple them into two separate systems and fuse them. So, the hand shape, orientation, and location will be addressed using state-of-the -art object detection model. As the vision algorithms are getting better every year, the Yolo algorithm meets the requirement of detecting at realtime with high

framerate and accuracy. For the motion trajectory to model this, we use the LSTM network because it has the sign language data short-term spatiotemporal feature and LSTM network shown [5] to model well for short time-series data. The system Depicts, A user who would wear a smartwatch on the wrist (wrist is the most involved EMA vowel finger). Since most people are comfortable wearing watches in daily life, we chose this platform for the study. The Inertial Measurement Unit (IMU) sensors (accelerometers, gyroscopes) on the smart watch are used for wrist tracking. The last part is fusing the result of those two models.

2.9 Related Works

A survey of several local and worldwide researches on sign language recognition is offered in this part. Many prior studies have attempted to design a system for recognizing and translating sign languages in general, and Ethiopian sign language in particular. Similar studies are being carried out to recognize non-Ethiopian sign languages. This is due to its importance for narrowing the communication gap that exists between hearing and hearing challenged individuals. The review focused on the research questions that we stated in this.

RQ1 In [27] thesis work in which they used Random Forests and KNN-DTW algorithms was used to recognize 18 Sexual health related words. For the data gathering, 3 times for each sign for DTW and from 4 signer to collect 3600 samples for the random forest of 6-point Kinect Xbox skeleton data. the first method they employ was DTW as feature extractor and KNN as classifier which the get result 76.25% accuracy. And with the second approach they received a accuracy 96.96% with Random forest. Thus, they conclude that the Random Forest have better performance than DTW-KNN. Their work was novel which address specific sexual health related word's but the reliability issues with data extraction device and device have limited mobility since it have to mount at specific height.

Tefera Gimbi devised a method for recognizing isolated signs in Ethiopian Sign Language,[4] The research was successful in achieving an overall system recognition rate of 86.9%. the research will focus on twenty-five signs. twenty of the twenty-five signs are one hand, the rest of the signs are done with two hand signs. each of the signers a sign is expected to be performed 20 times. They used 15 for training and the rest 5 for the purpose of testing. As a result, we will have 45 training videos and 15 instructional videos for each

sign. Video testing, for all 25 signs, they used a total of 1500 videos. The signers maintain a consistent tempo when doing the sign. in addition, this component is responsible for preparing the input for HMM training, the k-means technique was used to create cluster matrices for each symbol.

According to [28] the two authors sought to create a hand gesture identification system by combining the Gabor Filter (GF) with principal components Analysis (PCA) for featured extraction and ANN for identifying the ESL of 34 letters of the alphabet. The method employed white glove to capture the entire dataset to lessen the difficult of segmentation because the wide range of skin colors among the signers. Furthermore, non-ESL images were acquired from websites, and the system is capable to rejecting these non-ESL images, according to their research, The primary goal of this project is to convert the EMA alphabet signs into voice. The trials showed that with enough data, the ANN technique provided good results and quickly recognized an unknown input sign (non-EMA). According to the experimental data, the system has recognition rate of 98.53%[28]. However, because it could not convert basic and specialized Amharic letter signs to the matching alphabets, the study could not provide full translation functionality.

Abadi Tsegay [16]went on to expand on this research. For continuous Ethiopian sign language, he sought to design an offline candidate hand gesture selection and trajectory determination system. This recognition system may also determine hand movement trajectories by extracting candidate EMA frames from a video clip. Candidate Gesture selection (CGS) and hand movement trajectory determination are the two main components of the system (HMTD). For this module, the CGS combined two metrics: a speed profile of continuous gestures and a Modified Housdorff Distance (MHD) measurement, yielding an accuracy of 80.72 %. The HMTD uses angle, x- and y- direction to consider both hand gesture centroid from frame to frame, and it returned a result with an accuracy of 88.31%, furthermore, the overall performance of the system is 71.88 %. Except for the second, third, fourth, fifth, and sixth orders of Amharic alphabet signs, he did not focus on the first and seventh orders. He hasn't focused on translating these signs into writing.

Abudelhafiz et al. [29] have done research work titled 'translation of continuous sign in Ethiopia sign language to Amharic text'. The proposed system describes sentence and word level recognition using Random Forest algorithm. Using depth information for hand and body joints generated from Microsoft Kinect. Their work address movement emphasis and sign segmentation which appears in continuous sing language recognition and analyzing hand movement pauses. The data collected from the sensor is stored in gesture dictionary which contain 10 sentence and 9 words. Those gestured are matched using random forest. The performance the found accuracy of 84.4 % and 60 % at word and sentence level respectively.

The researchers created Amharic sign language recognition using the Amharic Alphabet sign in [1]. They concentrated on the EMA, which is made up of 33 different hand movements. The features of letter images are extracted from the source video first, and then the classification approach of SVMs and NN for recognition is used. This paper introduces Fourier descriptors and other multi-features. The 34 feature descriptors are calculated from each pixel of the entire image, while the others are calculated from small localized interesting images. Ten signers were expected to perform a sign 160 times for only four fundamental alphabet signs from among thirty-four first order Amharic sign language characters, according to the researcher. However, they choose 153 of the greatest ones. The results of the experiment with 18 classes of the EMA film showed the at features such as Fourier descriptors are simple, efficient, and effective in distinguishing hand forms. The system had recognition accuracy of 57.82% and 74 % respectively, finally, the SVM classifier was determined to be superior to the NN. Even though they had theoretical foundation, but Their recognition accuracy is low.

RQ2 to answer research question 2 most international researchers try to address the issue of sign language, gesture or human computer interaction. Since most of fields have share common problem that can be solved in similar manner. Thus, Vision based, sensor based or using both Vision and sensor in this section we have checked all methods that are used by scholar will be reviewed.

In [30] those researchers propose ArmTrack system in their Using an off-shell Samsung live gear smartwatch, they try to track the 3d posture of the arm i.e., both wrist and elbow. The system uses motion and magnetics sensor data. The design of 3D orientation of the wrist (which is determined via sensor fusion utilizing motion sensor and gyroscope) and the space of possible elbow location is constrained. They try to find a sphere around the shoulder points, then can further reduce search space from the elbow they called a point cloud. given the data, they can now estimate translation of motion of elbow through hidden Markov model. After elbow location is known, the wrist is calculated to shift along the direction of the wrist. They found the result 9.2 cm median error compare with Kinect sensor data as ground truth. Generally, in [30] work trace geometry of arm. Even though this work doesn't work for the specific gesture, it shows that we can track the motion of hand gestures, which is essential for our work. This only shows potential also their ground truth is an approximation.

Gesture recognition method with accelerometer and gyroscope using CNN and RNN layers. In addition to the researcher at [31], the work of Ji-Hae Kim et.al. presents deep learning-based gesture recognition using a motion sensor named deep gesture. This approach using a wrist-attached wearable band motion sensor to collected data they demonstrated the potential of proposed algorithms-based wearable sensor data is to archive a higher recognition rate for a broader range of gesture patterns and certain motions. In the system, they chose the peak variance of accelerometer and gyroscope data identified. After the selection, the sequence is searched for using a method of sliding windows to find the min and max values are detected. The min is set as a start and max value from the selected peak as the end of the gesture. The sequence of 256 data segments is based on 6-axis i.e., 3 axis accelerometers together with 3- axis gyro. The dataset defines nine gestures, and a total of 180000 data set of nine gestures was collected. This means 20000 for each gesture. For the proposed model and dataset, they found a result of 96%, 96%, 96% recall, precision and F1-score, respectively. This shows the potential of motion data with deep learning method for gesture recognition may provide a good score in accuracy, as stated in [31].

Among the international research reviewed. Jordan J. Bird et al. [1] [3]. Tir proposed system uses a camera as image for image collection and also contains VGG and MLP as

image feature extractors, the leap motion hand is used to record the numerical representation of hand. The system has the ability to recognize 18 BSL selected gestures using two deep neural networks are benchmarked and compared. Those are vision model by CNN and optimized ANN, and leap motion model by evolutionary search of ANN topology. Next, the two best models are fused for synchronized processing. Therefore, the dataset was collected from five different participants for thirty to fifty seconds. The results in an overall result of 94.44%. In addition to the main objective, supplementary features are learned. The hypothesis of this research is supported by the model's application to a collection of previously unseen data, which shows that a multimodality approach outperforms the single sensor method. Another finding of this study is when using transfer learning with wights learned through BSL, all three models (Vision, leap, and fusion) outperform standard random weight distribution in classifying ASL, with an accuracy of 82.55 %. The transfer learning multimodality approach was the best overall model of for ASL classification.

2.10 Summary

In this chapter, we made a review of some notable and recent research work done on different on sign language both local and international. Presenting all proposed methods in this thesis is beyond the scope. This is due to the goal that we want to achieve realtime dynamic sign language and, it is the most active research area different researchers have been publishing papers in different advancement in the area. Currently, there is not an efficient model to detect real time dynamic sing language.

Table 2. 1 Literature summary

Authors	**Paper title**	**Proposed method**	**Gap**
Yeseph F. et al. [27] (2010)	Ethiopian Sign Language Recognition Using Artificial Neural Network	Use ANN to recognize the ESL, The use PCA and Gabor filter as feature extractor.	Small dataset also they use white glove while collecting data.

Abadi Tsegay et al.[16] (2012)	offline candidate hand gesture selection and trajectory determination for continuous Ethiopian sign language	determine hand movement trajectories by extracting candidate EMA frames from a video clip. (CGS) and (HMTD).	In this research they try to track the hand movement. it does not perform well to vowels in EMA
Tefera Gimbi et al. [4] (2014)	Recognition of Isolated Signs in Ethiopian Sign Language	Use HMM for training and k-means for clustering. They used image extracted from 1500 mp4 videos.	Best feature extractors than method used in this research,
Abudelhafiz et al. [29] (2017)	translation of continuous sign in Ethiopia sign language to Amharic text	Used Random forest and Microsoft Kinect skeleton joint input.	System doesn't detect hand shape only motion
N. Kefyalew Tamiru [1] (2018)	Amharic Sign Language Recognition based on Amharic Alphabet Signs	classification approach of SVMs and NN. They use Fourier descriptors and other multi-features as feature extractor.	Their classification accuracy is low.
Getnet mesele [27] (2021)	Isolated word real time language recognition for sexual reproductive health using Kinect	Classification approach use DTW and RF. They use kinnect xbox skeleton data as input to detect 18 sexual health related words.	Reliability issues with data extraction device. The device is immobile and mounted as down side.
Ji-Hae Kim et.al. [30] (2016)	Deep learning based gesture recognition using a motion sensor	Gesture recognition method with accelerometer and gyroscope using CNN and RNN layers , to classify nine gestures	They Use simple gestures. But most of the gestures are complex in nature
Jordan J. Bird et al [3] (2020)	British sign language recognition via late fusion of computer vision and leap motion with transfer learning to American sign language	recognize 18 BSL selected gestures, using vision model by CNN and optimized ANN, and leap motion model then best models are fused for synchronized processing	Their model overfits. Since it preforms less to unseen data.

CHAPTER 3 : System Model and Description

3.1 Overview Of Methods

This chapter covers techniques and ways employed to model and implements Real-time EMA recognition the system architecture describes the overall workflow of the recognition.

The main objective of this thesis is to analyze image and sensor data for the sake of developing an accurate realtime EMA recognition system using vision and motion sensors to bridge the communication gap presented with the hearing-impaired community. The problem identified and proposed in this thesis work is the real-world problem, thus an empirical-based design science approach was chosen for this work. The methodology used to achieve the objective is summarized in Figure 3.1. since we have proposed a fusion of vision and sensor methods, the collection of datasets from trained professionals and volunteers is essential to train and test the performance of the proposed system. Since there Are no publicly available standard datasets. Preprocessing was done before applying these datasets in the proposed system, and these steps are mandatory to increase the accuracy of the overall proposed system by treating outliers and other misleading, unwanted instance issues. The performance of the fused system needed to be tested. Hence different performance evaluation metrics were used to measure the performance. The main methods in this thesis work are addressed under four subsections. In subsection 3.2, the dataset collection and preparation are explained, subsection 3.3 the data preprocessing and refining is explained, subsection 3.4 explains the architecture and algorithms of the proposed fused system. In the subsection 3.5 testing mechanisms like the standard performance measurement are selected to qualify the proposed models. Comparison's analysis of the fused model with existing. Section 3,6 and 3.7 the tools used for this work and experimental setup will be discussed.

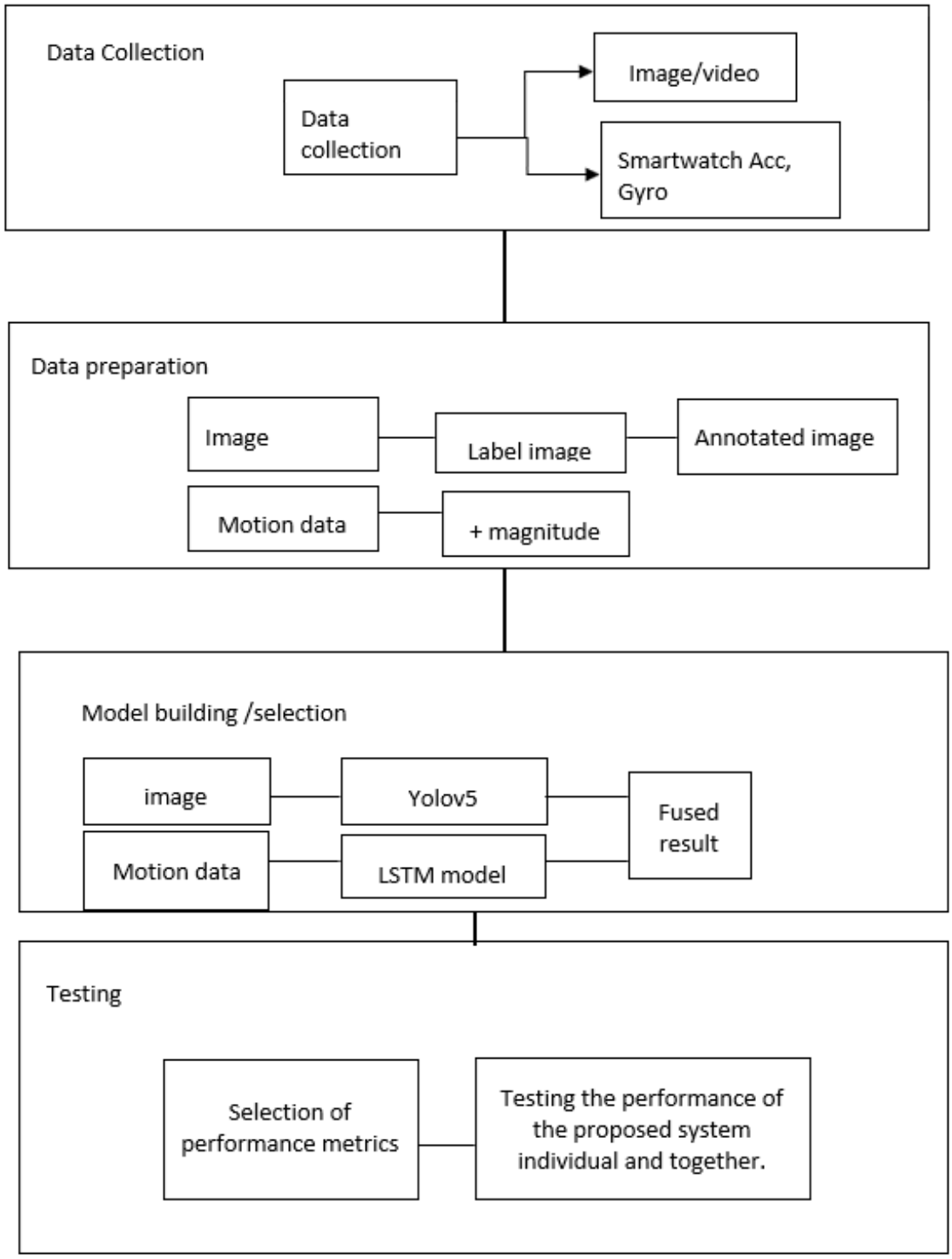


Figure 3. 1 General workflow of the thesis methodology

3.2 Data Preparation

In the domain of supervised machine learning methods (as in this thesis; classification) the data is mandatory [7]to test and validate the new model being developed with respect to the proposed problem to be solved as in this research, the prediction of EMA using vision and sensor motion data. To do so literature review of sign language and the associated existing work being accomplished in the domain are studied [1] [3] [31]. To have a detailed understanding of the behavior and attributes of the Ethiopian sign language, a sign language expert of this domain is consulted and discussed with them about the problem undergoing. About 2 sign language experts were considered during data collection, and they trained about 60 participants about the signing of the EMA alphabet.

3.2.1 Image Data

this vision part of the research requires a huge amount of image data. The easiest way to gather a large number of sign images, where through trained volunteers. Every volunteer was tasked to collect a 5-sec video recording while signing each EMA letter (33 In number). This collected video was converted to a frame of images. This way we were able to collect more than 250,000 EMA hand sign images. While collecting the image data since this was done using crowdsourcing, we were unable to use the same tool, whereas we ordered each volunteer to collect using their own-

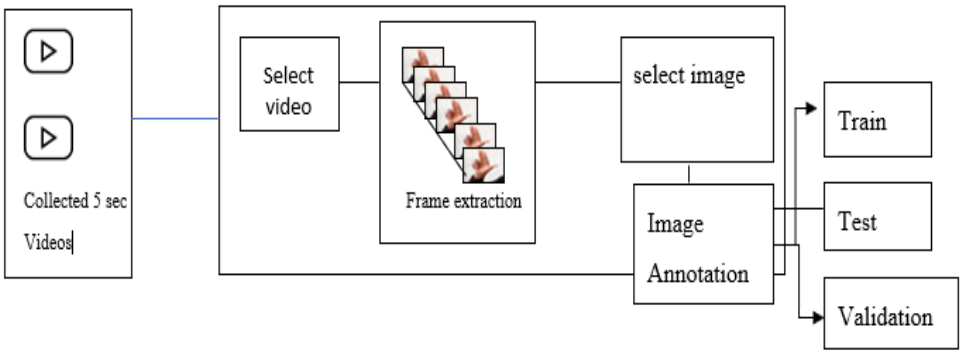


Figure 3. 2 The pipeline for mining dataset from video

smartphone at distance between .5m -1m from their camera. As expected from uncontrolled data collection, our data is not uniform it has a different background, different size, different distance, hand shape. Thus we were forced to collect large data so that we could select from rich size.

The images in Figure 3.3 are extracted frames of the videos recorded by the participant. Total 1980 video was collected (i.e., 60x30). The video was collected from April 2021 – June 30, 2021, in three-stage twenty participants at one time.

We have written a python script used to extract the video frames or convert the video into sequences of image frames. The script will save the images to a specific folder using the UUID python library for unique image naming.



Figure 3. 3 EMA hand sign (ኦ, ረ, ነ, ሠ, ሸ, ገ, ኸ, ደ, ጨ) respectively starting top left

During data collection other than the duration of the video, we do not constrain what they should wear or when to collect. Thus give the data to rich in variation. The image has different sizes, background, hand size, color etc.

As recommended by [32], for the best result, they recommend >=1500 images, so we did follow as they recommended. Thus

Table 3. 1 summary of EMA image dataset, we use total images 49500 images

Number of classes	Per class	Train 80%	Val 10%	Test 10%
33 class	1500	39600	4950	4950

3.2.2 Motion Data

In the other part, which is sensor data, as in [31], [33], [34], in this research work, we used a smartwatch. Thus sensor data was collected from 3 other trained volunteers while wearing a smartwatch.

In this research, we use TicWatch 2 watch (with android Wear OS). during the selection of watches, we emphasize two main points: first, its operating system must be android wear. Thus we can build an app that listens to accelerometer and gyroscope sensors. Secondly, the watch must have Wi-Fi, the main reason to achieve realtime detection, and Wi-Fi is faster than Bluetooth communication. We must choose fast communication means to send the watch sensor data from device to pc. With the two-reason also mentioned chip price we choose TicWatch 2.

To collect the motion data from a smartwatch, we have to build

- Client application – that is an android wear app that sends sensor data from the watch. We build this app using flutter. Send sensor data to provided IP at the required time.
- Serve application – that listens to data sent by the smartwatch and saves it in the form of CSV. We program the serve using python Django as backend. It listens to specific port 8000 in this case to receive incoming data.

We build the motion data set of EMA vowels, and we ask three participants to perform 7 gestures (movements). In controlled environment they have to sit in chair. And perform the 25 reputations of 1 sec long movement at time. As shown in Figure 1. 1

The gesture to be performed

1. Geez – stationary pose the hand shape without movement only pointing your palm to the other person.
2. Kabee – move your hand while posing to the left.
3. Sals – move your hand while posing to right
4. Rabee – move your hand while posing downward.
5. Hams – move your hand while posing, circular clockwise, also your hand must also maintain to show your palm to whom you are speaking.
6. Sads – move your hand while posing, zigzag downward.
7. Sabe – move your hand while posing, flip your hand clockwise horizontally,

We collected 7*500*3 data (10500 data), seven for vowels, 500 data collected from each person, three number of participants, and data were collected at a sampling rate of 60 Hz and transmitted to the smartwatch. One window of data is 1 sec long.

The following features are collected from the smartwatch

- Acceleration (X, Y, Z)
- Gyroscope (X, Y, Z)

In addition to the three-axis Acceleration and Gyro data, 4th parameter was added, this is the magnitude of the three axes Acc and Gyro after data collection.

The Euclidean norm

$$||Am||_2 = \sqrt{A_x^2 + A_y^2 + A_z^2} \qquad (4.1)$$

Thus, our data has eight parameters Ax, Ay, Az, Am, Gx, Gy, Gz, Gm. x, y, z are the axis and the m subscript refer to magnitude.

3.3 Data Preprocessing And Labeling

image preprocessing is one of the major tasks in image processing. but yolov5 model does not require much preprocessing when compared to traditional machine learning since it has the ability to work on raw data

Image labeling graphical annotation of images. In object detection, the input image dataset must be annotated, which means the area of interest in the image must be selected so that the model will recognize the area of interest. In own dataset, each input image are annotated by drawing a bounding box close to the hand shape.

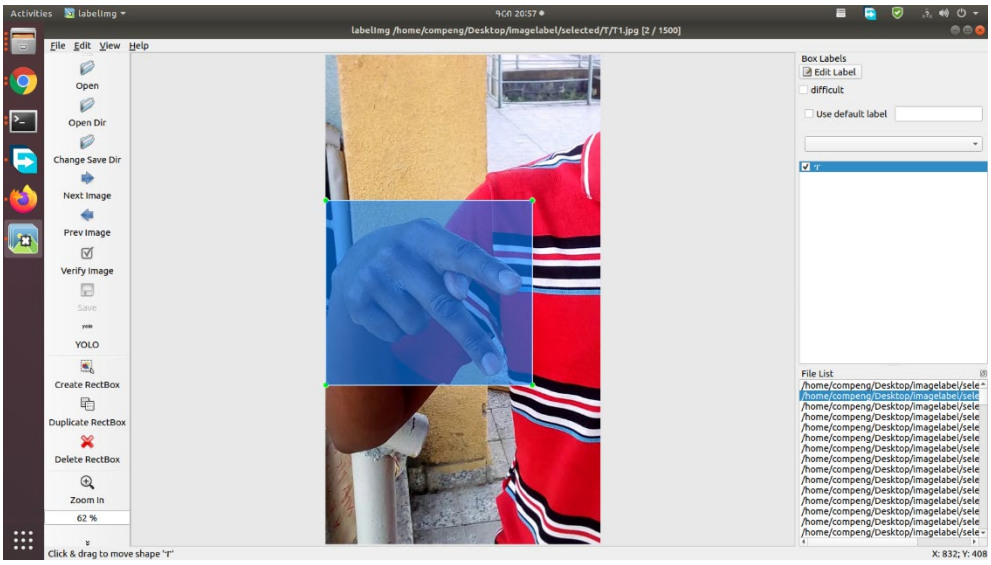


Figure 3. 4 labeling image, area of interest which is the hand shape of letter T, this is done using LabelImage python package. There are allot of free image labeling tools but LabelImage is open source and easy to use labeling tools.

3.3.1 Data Augmentation

Researcher [32: There are various ways to improve the model's performance without increasing the inference time lag. Because they cannot change the model's inference time, most work on data management and augmentation in the training pipeline. These

techniques supplement and broaden the training set, exposing the model to circumstances that would otherwise go unnoticed.

The data augmentation that yolov5 uses is employed during the training. Thus we didn't need to perform augmentation because of two reasons first, we prepare enough training data, and lastly, the model does augmentation while training, those are augmentation mechanism used by yolov5,

- Cutouts: Cut elements of one image and paste them onto the augmented image to create a composite image. The model must learn to generate predictions based on many features due to the picture cutouts [26].
- Mosaic data augmentation: combines four training images in a specific ratio to make a single image (instead of only two in CutMix)[32].

3.3.2 Data Visualization

A graphical depiction of information and data is known as data visualization. Data visualization tools make it easy to examine and comprehend trends, outliers, and patterns in data by employing visual elements like charts, graphs, and maps. Let explore what the sensor data look like. Those images below show Ax, Ay, Az, of accelerometer also Gx, Gy, Gz of gyroscope reading from a smartwatch at 50Hz.

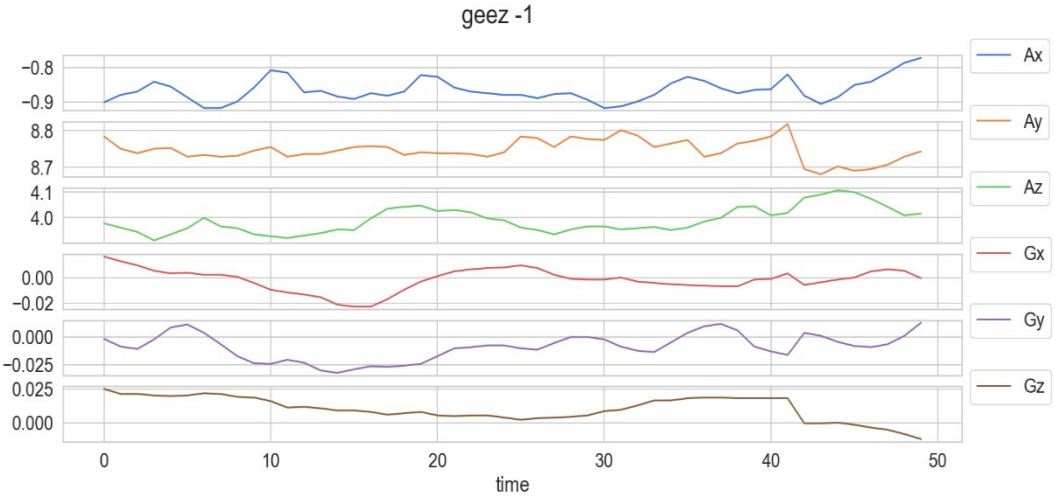


Figure 3. 5 one frame of geez (the first vowel of EMA)

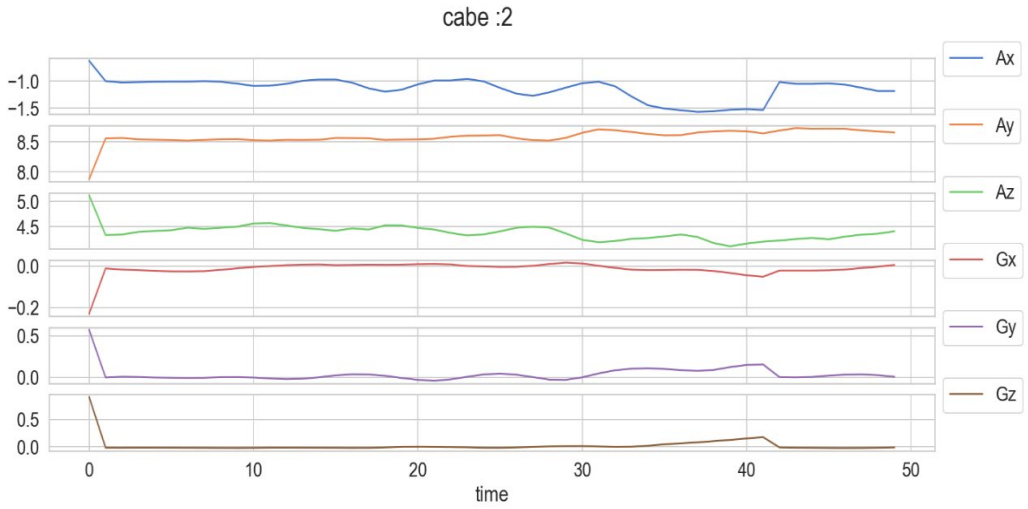


Figure 3. 6 frame of the cabee (the second vowel of EMA)

Those Figure above present what our smartwatch sensor data looks like. The parameters plotted are Ax, as accelerometer x in m/s^2. The same for Ay and Az. The Gx, Gy, Gz are three-axis gyros in DPS.

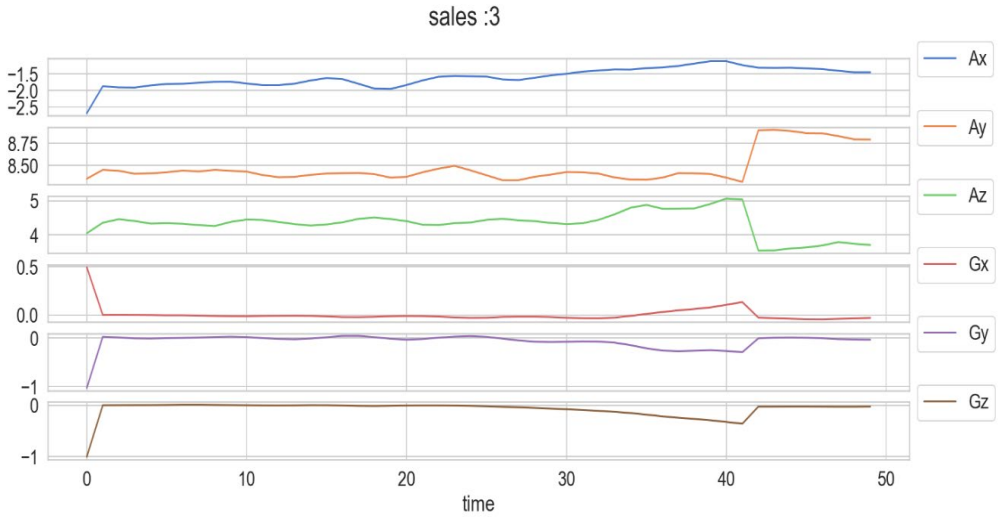


Figure 3. 7 frames of the sales (the three vowels of EMA)

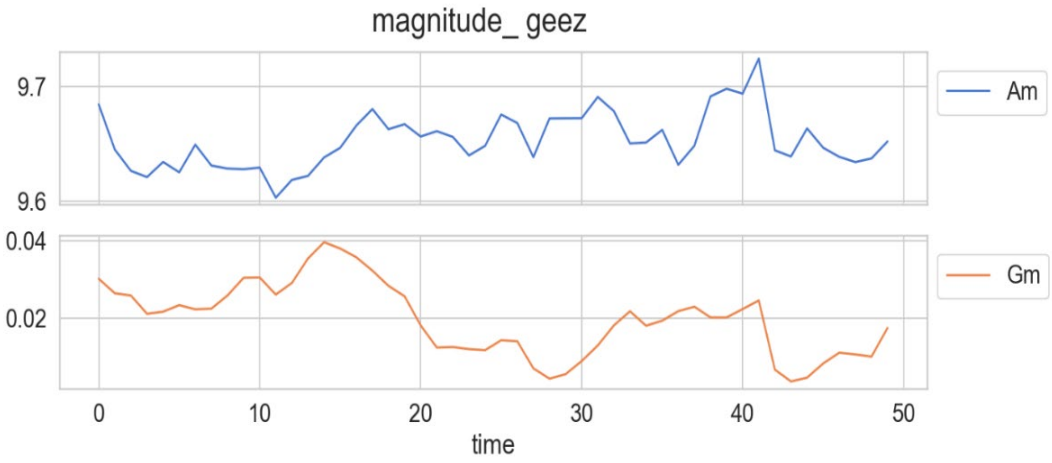


Figure 3. 8 magnitude feature of geez the first vowel and

As we can see from the fig, fig the plot looks the same, thus our model my get confused and we add other feature in addition to 3 axes gyro and accelerometer that is the magnitude of both reading the Figure have.

In the magnitude feature, we can see that the two have differentiating factors, and adding this feature helps our model to distinguish the vowels clearly. The see in Figure 3.5 -3.7. the data does not show much change due to the span it took for the EMA vowel to perform. The addition of the magnitude feature, as we can see from Figures 3.8 -3.10, the magnitude can support our models for better classification.

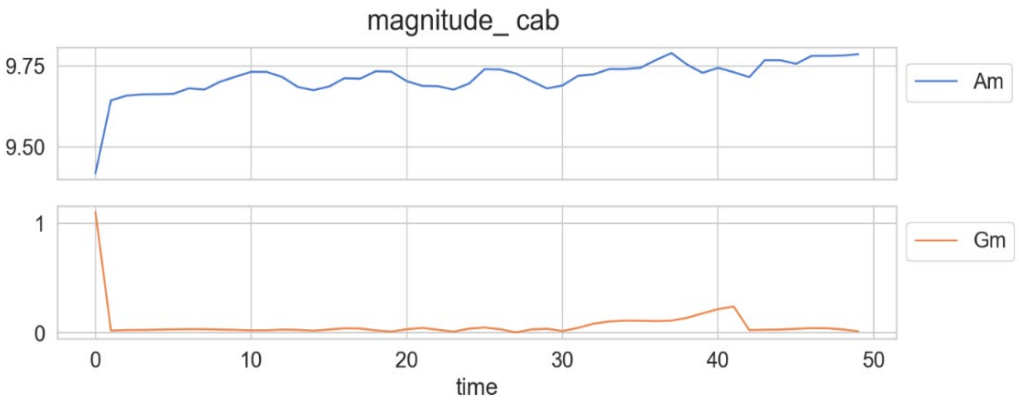


Figure 3. 9 magnitude feature of geez the first vowel and cabee

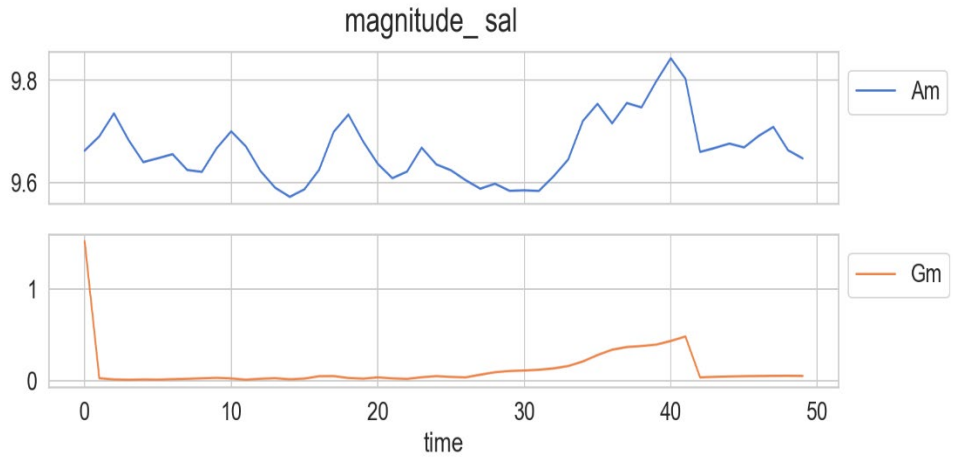


Figure 3. 10 magnitude feature of salse the third vowel

3.4 The Proposed Model

The proposed model tries to combine the two methods: the vision method using yolov5 and the device method using smartwatch sensor motion data. Those models are developed to perform two different tasks the vision method to detect and classify the hand shape and orientation of EMA hand sign. Moreover, the device method to detect and classify to hand movement of EMA. Finally, the fused model will merge the to result for complete realtime detection.

3.4.1 YoloV5 model

Yolo [35]is a convolutional neural network (CNN)-based object detection system that produces rapid and accurate results. The YOLO (you only look once) strategy is when you look at something once thus, the neural network only sees the input image once. It also anticipates the images observed object. It operates based on splitting the input image into several grids based on a predetermined grid size and predicting the likelihood of each grid. In each grid, the needed object it foresees all of the classes. The item boundaries that appear in the run of the image. As a result, it became a rapid end-to-end object detection. It can be used for realtime detection as well. Yolo algorithm is being improved continuously in [26], [31]–[34]

Its parameter quantity occupies most of the parameter quantity of the entire network. By adjusting the width and depth of the BottleneckCSP module, four models with different parameters can be obtained, namely YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. The SPP module mainly increases the receptive field of the network and acquires features of different scales. YOLOv5 also adds a bottom-up feature pyramid structure based on the FPN structure. With this combination operation, the FPN layer conveys strong semantic features from top to bottom, and the feature pyramid conveys robust positioning features from the bottom up. Combining feature aggregation from different feature layers improves the network's ability to detect different scales' targets. In this research work, we choose YOLO v5s since they have good performance as the other version yolo, relatively small with 7341398 parameters.

Note: to train the EMA using yolov5, and the code is available in their ultralytics github repo. We did not write our code, we use their code to train our own custom dataset EMA images.

3.4.2 LSTM Network for modeling time series

Data from time series can be used to confirm trends in data from the past and present[38]. Time series data is also sensitive to time-based information. Time series data is used in various fields to solve classification problems. Selecting a good classification model is essential. Dynamic time wrapping, machine learning techniques [33], and shapelets were developed to solve the time series classification problem.

LSTM has been successful in many sequences learning and temporal modeling tasks. [39] [40], [41] such as speech, recognition, handwriting recognition, and sentiment analysis etc.

The motion trajectory in EMA in this research, we used IMU sensor (gyro, acc) data from smartwatch to model motion, thus making it the data time series. Many researchers have used different methods to model time-series information. Lastly, LSTM has been state of the art in modeling time-series data, with their success in a seq-seq model in word embedding [42] showed the effectiveness of LSTM in modeling shot temporal information

as in our case [43][44] . In this research work we will explore to LSTM models name vanilla LSTM, simple one hidden unit LSTM network and Bi-LSTM.

3.4.3 Fusing The Models

A single modality technique is used in much of the state-of-the-artwork in Sign Language recognition, with multimodality studies being some of the most recent works in the area [3], [43]. Since we decouple the task in two different models: vision (using yolov5s) and motion (using LSTM network). We fuse the model by simply combining the prediction result of both models. Since both correspond to different perdition which is the yolo to hand shape, whereas the LSTM to motion trajectory, thus, will give as EMA recognition from both.

3.5 Evaluation

To evaluate and qualify the proposed method, yolov5 model, LSTM model, and the fused of both models, different performance metrics have been used, such as accuracy. Recall, F1-score, precision, most of those metrics and extracted from the confusion matrix. And subject to the performance of the classifier. Several common terms to be knowns are:

True positive (TP): predicted positive and are actually positive.

False positive (FN): predicted positive and are actually negative.

True negative (TN): predicted negative and are actually negative.

False negative (FN): predicted negative and actually positive.

Selected performance evaluation parameters are explained as follows.

Accuracy: The measure of how the model classify the given data correctly.

$$\text{Accuracy} = \frac{(TP+TN)}{(TP+FP+TN+FN)} \qquad (3.1)$$

Recall: The percentage of positive instances out of the overall actual positive instance. So, denominator (TP+ FN) is here the actual number of positive instances in the dataset. Meaning "how many more correct ones, the model missed when the right ones were seen."

$$\text{Recall} = \frac{TP}{(TP+FN)} \quad (3.2)$$

Precision: The level of positive instance out of all predicted positive cases. Here denominator is the model forecast done as positive from the entire given dataset. Meaning "how much the model is correct when it says it is correct."

$$\text{Precision} = \frac{TP}{(TP+FP)} \quad (3.3)$$

F1- Score: It is the harmonized mean of precision and Sensitivity. This takes the commitment of both, so higher the F1 score. The better. Because of the item in the numerator on the off chance that one goes low, the last F1 score goes down essentially. So a model does well in the F1 score if the positive predicted are positives (precision) and doesn't missed out positives and predicts them negatives (recall).

$$\text{F1 score} = \frac{2*Precison*recall}{(Precision+recall)} \quad (3.4)$$

Mean Average precision

Average precision mostly used to measure accuracy in object detection like Yolo. The mAP score is calculated by taking the mean Ap overall classes and overall intersection over union thresholds. For our experiment we will intersection mPA_0.5 as accuracy to match without LSTM network.

$$mPA = \frac{\sum_{i=1}^{k} Ap_i}{k} \quad (3.5)$$

3.6 Hardware And Software Tools Used

3.6.1 Hardware

To capture sensor data from participants TicWatch 2 Smart watch was used. And to design and implement the proposed model using software a computer with specification of CPU intel® core i7-3770 CPU @ 3.4 GHz with 12GB RAM (internal memory). To train the yolo model, Colab is a Google Cloud Platform integrated development environment (IDE) for executing Jupyter notebooks (GPC). Colab is a free service that gives you a temporary

Linux instance with access to a single Tesla P100-PCIE-16GB GPU via a Jupyter notebook. We use the Pro-version of Google Colab since we need high ram, and uninterrupted run. this have up to 24 GB Ram and 24-hour time limit.

3.6.2 Software Tool

To implement this research phyton is used as a programming language and PyTorch and Keras as a higher-level application programming interface (API) which can found as a package in anaconda development environment.

3.7 Experimental Setup

In this thesis work, the performance of the yolov5 model which is trained from scratch and LSTM network is evaluated

3.7.1 Hyperparameters Setup

Hyperparameters help a model to give significant results choosing the right value of hyperparameters will result in getting an accurate model. But there is no predefined rule to set hyperparameters or there are no fixed values for hyperparameters to set for learning algorithms therefore experiments should be conducted and take values that give good results [37]. So, for this research, different hyperparameters value tuned and selected. Form different hyperparameters number of epochs, learning rate, dropout rate, batch size, loss function, activation function and optimization algorithm were selected and optimized. Finally, the result shows that the optimized hyperparameters were given a remarkable result.

- **Number of Epoch:** It defines the number of times that the learning algorithm will run through the whole training dataset has the opportunity to update the internal model parameter. Small number of epochs lead the model to be underfit (cannot learn enough, both test and training set accuracies are low) and if too many epochs are used, the model might over-fit (model memorizes the training samples). If the model is overfit the training accuracy will be high and test

accuracy will be low. For this research 25 and 50 epoch is used and the model performance is evaluated at each epoch.

- **Learning Rate:** Learning rate is used to update weight in the training process. It controls the amount of weight to update during the backpropagation process. Choosing an efficient learning rate is difficult due to using a small learning rate makes the model train slowly, and also the model became optimal and if the learning rate is large the model will train fast, but it will not be an optimal model. In this research different learning rates were used, and the optimal value was selected which is 0.01.
- **Dropout rate -** It randomly discards neurons to prevent overfitting problems based on the given value.
- **Batch size:** Defines the number of input data that passes into the model at a time. This helps the machine to process data by breaking input data into several small batches instead of providing all data at the same time. So, this helps the machine to train fast or reduce computational time. In this research, batch size of 16 32, 64 is used and evaluated.
- **Loss function:** In the Machine learning process, we need to minimize the error made in training this can be performed by optimization algorithms. Loss function can be selected based on the problem that we work on whether classification or regression. In this research the type of problem that we are working on is a multi-class classification. Therefore, a cross entropy loss function is most widely used loss function for multi class classification problem. The other factor that leads to select cross-entropy loss function is the output layer activation function, Softmax. An activation function also another factor to select loss function. A cross entropy loss function computes the cross-entropy loss between the input label (actual class output) and predicted output probability of the model. Then the result is calculated, and the difference is measured to optimize the loss.
- **Activation function –** In the output layer Softmax used as activation function since we have a multi-class problem. Softmax is used as a classifier instead of

random activation since Softmax can normalize it input value into probabilistic distribution where it summed value gives one.

- **Optimization algorithm:** there are different ways of calculating weight in supervised learning one of the widely used method is backpropagation algorithm. In this algorithm the optimization starts from output and propagate backward and update the weight. In this research, the optimized gradient descent (SGD) is used. SGD is an iterative method which used to optimize the loss function. It is a stochastic approach of gradient descent instead of calculating gradient descent over the entire data. It calculates on randomly selected subset of the data. This reduces the computational time and complexity.

Table 3. 2 Hyper parameter values

Hyperparameters	**Models**		
	Yolov5 -s	**Bi-LSTM**	**Vanilla LSTM**
Number of Epochs	25	50	50
Batch size	32	16	32
Loss function	Cross Entropy	Cross Entropy	MSE
Optimizer	SDG	Adam	Adam
Activation	Softmax, leaky ReLU	Softmax, ReLU	Linear, Softmax
Learning rate	0.01	0.01	0.001
Dropout rate	-	0.5	0.2

CHAPTER 4 : Result and Discussion

This chapter tries to describe the experimental results of the proposed modified model. In this section, experiment results were briefly discussed. Also, a comprehensive set of experiments performed to evaluate the individual performance of the proposal model.

4.1 Experimental Results

In the experiment stage EMA detection and classification is performed in two ways the first one is by using the YoloV5 model, and the second one is the LSTM network. Also, different hyperparameters are optimized to get accurate classification results. Like other training processes, the data split into three for training, validation, and testing. Let us see each model result in detail.

4.1.1 EMA Classification Using Yolov5 Model

For this research, the Yolov5 model was used to classify and recognize Ethiopian manual alphabet. The proposed model accepts 416x416x3 input images and every input is reshaped to the specified size. and finally gives 33 category results. The model is trained with 39600 Images and gives a remarkable result. Since the Yolov5 model requires a large number of images for the training process. To get an accurate model, different experimental scenario setup were used and conducted.

4.1.1.1 Result using different learning rate

The result obtained from using different learning rate value is described in the table below. To measure the performance of the model, metrics such as accuracy (mpA@0.5), loss (classification loss), precision, recall and F1 score are used. These metrics applied to the training, validation, and testing data. For this scenario, all hyperparameters are the same except for the learning rate.

Hyperparameter setups

- Number of epochs = 25
- Optimizer = SGD (Stochastic gradient descent)
- Activation = *Leaky ReLU*
- Learning rate = 0.01
- Batch size = 32

Table 4. 1 experiment result of yolov5s using different learning rate

Learning rate	Mean accuracy			Mean loss		weighted precision	weighted. recall	weighted F1
	Train	Val	Test	Train	Val	Test	Test	Test
0.01	96.2%	97.5%	96.2%	1.3%	1%	96%	99%	97%
0.001	50%	48%	62%	7.4%	5.2%	50%	64%	56%

The model gives a different result for different learning rates for the first learning rate value the model has less mean loss value but the model's ability to generalize for new data is poor. For second learning rate, the mean accuracy for training, validation, and testing are high also weighted precision, recall, and F1 score are best when compared with remaining two learning rate value but the mean loss is a little bit higher than first learning rate value. So, based on Table 4.1 we can say that the third learning rate value is an optimal value.

4.1.1.2 Result using different Batch size

The second experimental scenario is using different input batch sizes for the training, validation, and testing process

Hyperparameter setups

- Number of epochs = 25
- Optimizer = SGD (Stochastic gradient descent)
- Activation = *Leaky ReLU*
- Learning rate = 0.01

Table 4. 2 Experimental results of the yolov5 model by using different batch size

Batch size	**Mean accuracy**		**Mean loss**	
	Training	**Validation**	**Training**	**Validation**
32	96.2%	97.5%	1.3%	1%
64	78%	82.5%	3%	1.9%

Test set				
Batch size	**Accuracy**	**weighted Precision**	**weighted Recall**	**weighted F1 score**
	Test	**Test**	**Test**	**test**
32	96.2%	96%	99%	97%
64	74%	84%	81%	81.9%

For the smaller batch size, the model performs well on each evaluation metrics since in the smaller batch size the model starts to learn without seeing all the data at once. So, the ability of the model to generalize is good. From the above Table 4.2 we can generalize that smaller batch size is optimal for the proposed model.

4.1.1.3 Result using different optimizer

For this research, three different optimizers are used to compare the best performing one and the result is described in the below table by applying various classification accuracy metrics on both training data (train and validation) and test data. For this scenario, all hyperparameters are the same except the optimizer function.

Table 4. 3 Experimental results by using different optimizer function

Optimizer	**Mean accuracy**		**Mean loss**	
	Training	**validation**	**Training**	**validation**
SGD	96.2	97.5%	1.3%	1%
ADAM	46%	67%	5.9%	4.8%

Test set				
Optimizer	**Accuracy**	**mean Precision**	**mean Recall**	**mean F1 score**
	Test	**Test**	**Test**	**test**
SGD	96.2	96%	99%	97%
ADAM	74%	62%	81%	70.23%

From the above tables, it can be clearly seen that SGD's performance is the highest with training accuracy of 96.2%, validation accuracy of 97.5 %, and test accuracy of 96.2%. Generally, the proposed model makes predictions for different EMAs by taking image inputs, it scores 96.2% mean training accuracy, 97.5% mean validation accuracy, and 96.2 test accuracy with learning rate 0.01, batch size 32, data split ratio of 80:20, and SGD optimizer function.

4.1.1.4 Result analysis for the selected model

Since the proposed model trained from scratch and also, In the 10th epoch both training reached 89%. In between the 10th epoch to the 20th epoch training accuracy slightly increase over time. In the final epoch training accuracy reached 96.2 %. this can be shown in Figure 5.9. On the other hand, when we see a model loss at the beginning the training

and training loss were high it scores around 0.08 but after the fifth epoch, it decreases linearly from 0.08 to 0.40 for training. In between 10th to 20th epoch both the training loss decreases from 0.40 to 0.018. In the final epoch or from 20th epoch to 25th epoch loss in training reduced to 0.0013 classification labeled.

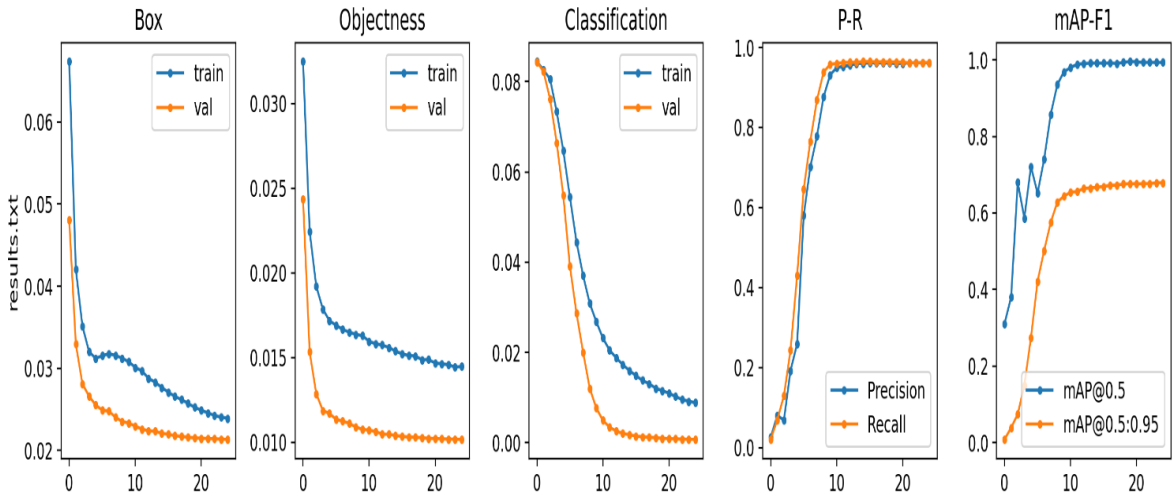


Figure 4. 1 overall result plot of yolov5s

Since most object detection model use mean average precision and mAP is measuring accuracy of object detection. The yolov5 repository give as rich metrics like box loss, which measure the error in boundary box of the detected object and ground truth i.e., bounding box accuracy. As we can see in the Figure 5.9 box loss decreasing at each epoch this convey that our model localizes the box with 3% loss. The objectness loss the confidence of measuring the objectiveness of box. And the classification loss of class conditional probabilities.

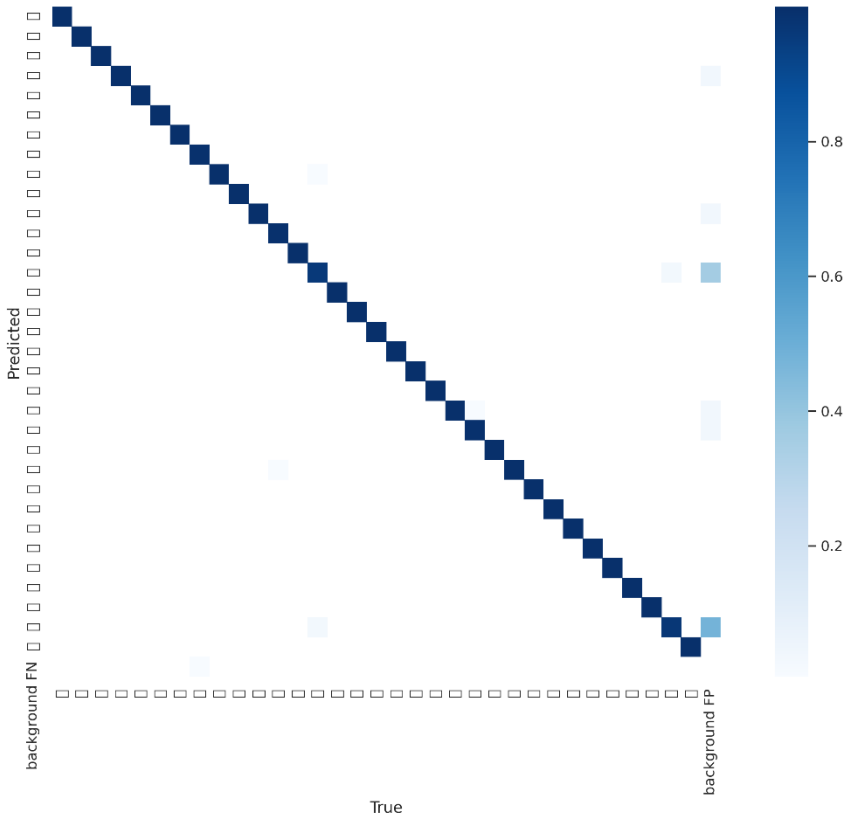


Figure 4. 2 Confusion matrix of selected model

The confusion matrix in this ‘ሀ’,’ለ’,’ሐ’,’መ’,’ሠ’,’ረ’,’ሰ’,’ሸ’,’ቀ’,’በ’,’ተ’,'ቸ','ኀ','ነ','ኘ','አ','ከ','ኸ','ወ','ዐ','ዘ','ዠ','የ','ደ','ጀ','ገ','ጠ','ጨ','ጰ','ጸ','ፀ','ፈ','ፐ’ respectively. this due to the matplotlib does understand the niyal.tff geez the letter encoding.

On the other hand, figure 5. Shows the confusion metrics which indicated that the capability of the model to generalize for new data is good.

Finally. The overall performance of the model can be summarized in the table below by using different evaluation metrics

Table 4. 3 model overall performance

Metrics	**Mean accuracy**		**Mean loss**	
	Training	**valuation**	**Training**	**validation**
Score	96.2%	97.5%	1.3%	1%

Test set				
Metrics	**Accuracy**	**weighted Precision**	**weighted Recall**	**weighted F1 score**
	Test	**Test**	**Test**	**test**
Score	96.2	96	99	97

4.1.2 LSTM network evaluation

In order to decide with the motion, we consider two types of LSTM network algorithms such as vanilla LSTM, Bidirectional LSTM to build this study. The performance accuracy of these models was evaluated on the dataset for the class EMA vowels.

Evaluation vanilla LSTM sensor data

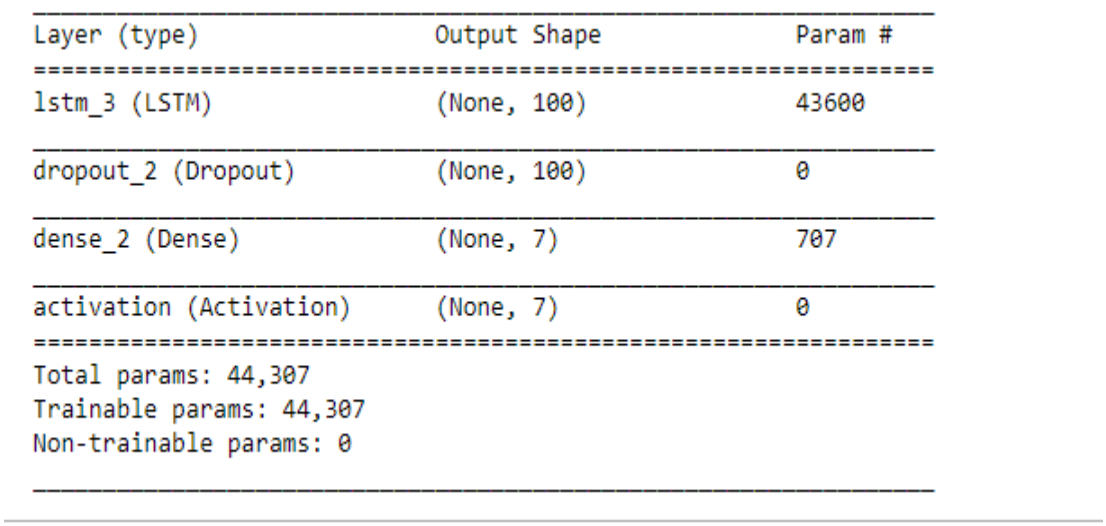
The first thing to do is select the proper hyperparameters using the parameters selection method, which is called the Keras hyperband tuner method, chosen and used among others. From this method, the proper model trainable parameters were selected. We use a moderately sized vanilla LSTM network which contains an input layer, hidden layer and

output layers. Finding a suitable batch size, a number of epochs, drop out function, dropout rate, activation function etc.

To find best hyperparameters we use Keras tuner, build a model using hypertuning, we define hyperparameter search space that is showed in the table above link in addition to the model architecture that can be found in the Fig summary The table shows the important parameters and its set values using a Keras tuner method, these are to implement and experiment all the model.

Table 4. 4 list of Implementations hyper parameters and it's set of values

Parameter	Keras tuner settings	Setting values
Number of hidden units	8,60,100, default 8	100
Batch size	16,32,128 defaults=16	16
Number of epochs	Max 50	50
Optimizer	Adam	Adam
Activation function	ReLU, tanh, sigmoid	ReLU
Loss function	MSE, categorical cross entropy	MSE
Drop rate	Min =0.0, max =0.5 Step =0.005 default =0.01	0.2



Layer (type)	Output Shape	Param #
lstm_3 (LSTM)	(None, 100)	43600
dropout_2 (Dropout)	(None, 100)	0
dense_2 (Dense)	(None, 7)	707
activation (Activation)	(None, 7)	0

Total params: 44,307
Trainable params: 44,307
Non-trainable params: 0

Figure 4. 3 model summary vanilla LSTM

In the training phase, the model was evaluated by using training accuracy and loss. Since the model trained with other data the weight is optimized. So, for our data in the first epochs the model got training accuracy of 29% and it increases gradually to 92% at the 7th epoch. From the 10th to 20th epoch the model reaches a training accuracy of 94.5%. The model starts to decrease loss and increase the accuracy at each epoch. On the other hand, the validation accuracy in the first 10 epochs the model achieves 90% of validation accuracy finally it only reaches 91% of validation accuracy. Based on this we can say that the model suffers. Form a high percent of overfitting. In general, we can clearly see what discuss above in the figure below.

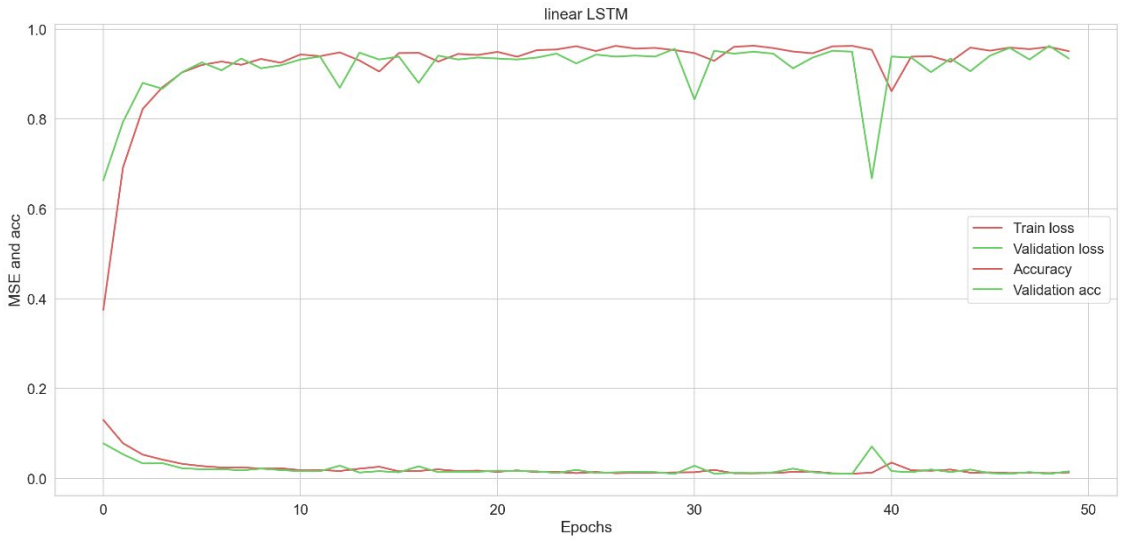


Figure 4. 4 Training and validation, accuracy and loss

Table 4. 5 vanilla LSTM network performance

model	Mean accuracy			Mean loss			Ave. precision	Ave. recall	Ave F1
	Train	Val	Test	Train	Val	Test	Test	Test	Test
Vanilla LSTM	92%	91%	93%	3%	2%	1.6%	94%	93%	93.4%

Evaluation of bidirectional LSTM

The following captures and table describe the experimental result of the Bi-LSTM model for EMA motion sensor data. The network has one layer, a hidden layer with 128 LSTM units, a dropout layer with 0.5, and 2 dense layers. we define the hyperparameter search space that is shown in the table below and the model architecture found in the Figure 4.5 summary.

The table shows the important parameters and its set values using a Keras tuner method, these are to implement and experiment all the model.

Table 4. 6 lists of Implementations hyperparameters and it's set of values

Parameter	Keras tuner settings	Setting values
Number of hidden units	8,60,100, default 8	128
Batch size	16,32,128 defaults=16	16
Number of epochs	Max 50	50
Optimizer	Adam	Adam
Activation function	ReLU, tanh, sigmoid	ReLU
Loss function	MSE, categorical cross entropy	MSE
Drop rate	Min =0.0, max =0.5 Step =0.005 default =0.01	0.5

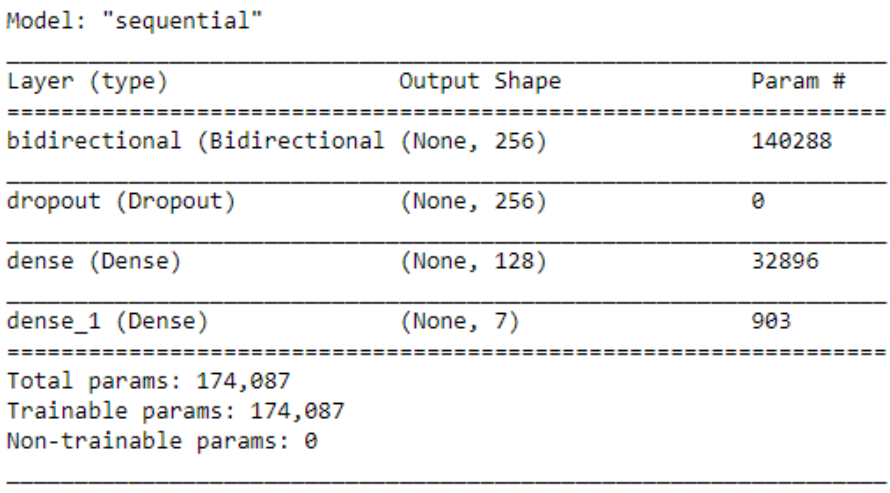


Figure 4. 5 capture of configuration of the proposed Bi-LSTM model

During training the model evaluated by using training accuracy and loss. So, for our data in the first epochs the model got training accuracy of 49% and it increase gradually to 93.56% at the 10th epoch. From 10th to 30th epoch the model reaches training accuracy of

97%. In the 40th epoch the model reaches 98% training accuracy then decrease to 97.56% until the end. On other hand when we see validation accuracy in the first 10 epochs the model achieves 91% of validation accuracy. Finally, it only reaches to 95% of validation accuracy. In general, Figure 4.7 and Figure 4.8 show the above discussed results and confusion matrix.

Table 4. 7 Bi-dictional LSTM network performance

model	Mean accuracy			Mean loss			Ave. precision	Ave. recall	Ave F1
	Train	Val	Test	Train	Val	Test	Test	Test	Test
Bi-di LSTM	93%	92%	96%	13%	26%	15%	94%	95%	94%

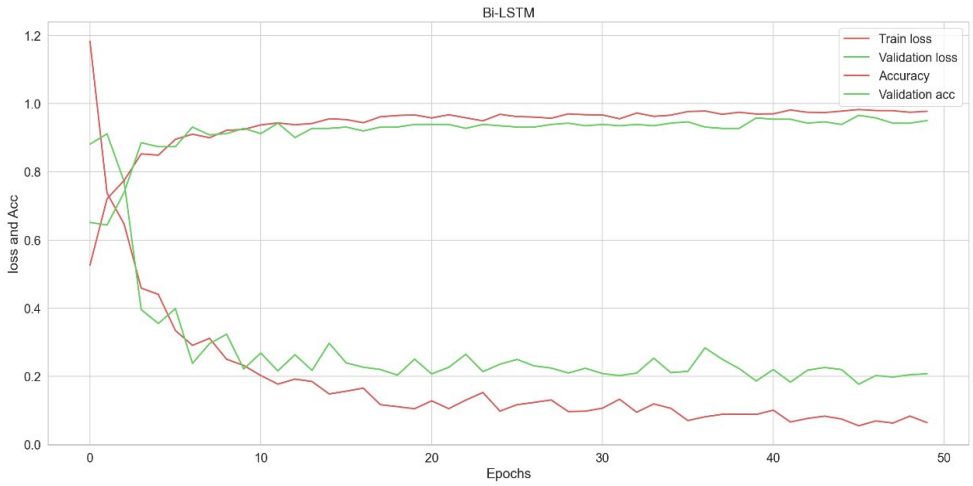


Figure 4. 6 Training and validation, accuracy and loss Bi-LSTM

The model performance on the testing dataset is more important than performance on the training dataset because, the new testing data record measures how the model can generalizes what it trained and respond to the new record.

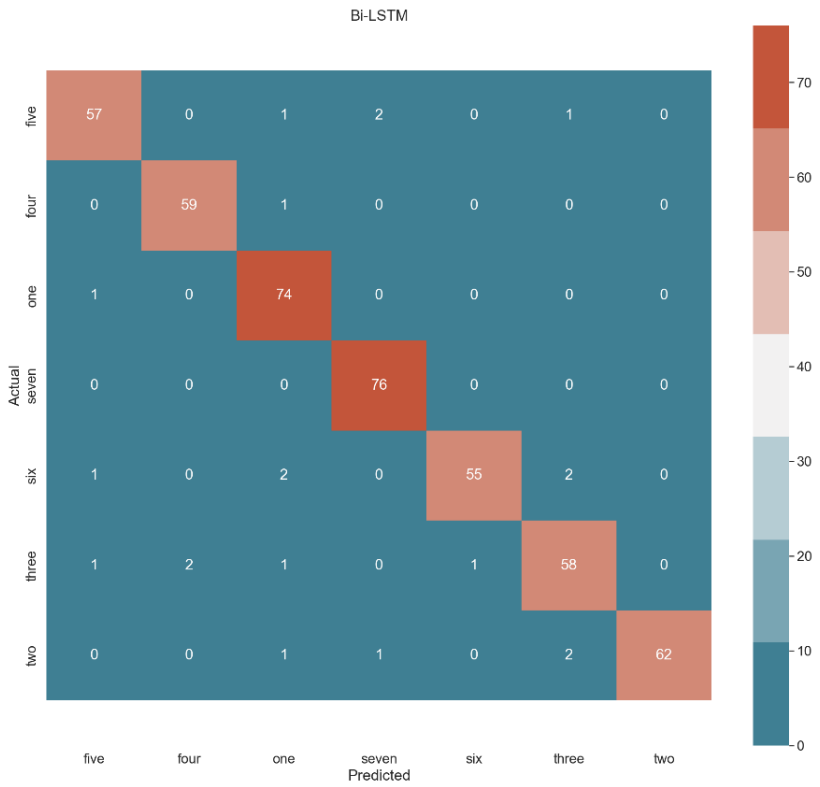


Figure 4. 7 Confusion Matrix of Bi-LSTM

Confusion matrix is a well-known evaluation Metrix for classification. Hence the confusion matrix visualization Bi-LSTM on testing data sample is computed and shown in Figure 4.7

4.2 Discussion

In the previous section, we have seen different experimental procedures using different LSTM networks, vanilla LSTM, and Bi-directional LSTM from motion trajectory modeling. And yolov5s for hand shape (vision) model. All the experiments were conducted with stated hyperparameters and hardware. To compare the best model, different evaluation metrics were used based on the evaluation metrics.

Form table 6,8,10, the mean training accuracy of the yolov5s model, vanilla LSTM and Bi-LSTM is 96.2 %, 92% ,93% and also the mean validation accuracy of model is 97.5%, 91%, 92% respectively. For the yolov5 model and vanilla -LSTM mean training accuracy and mean validation accuracy have 1.3%, 2%, respectively, we can say that the models do

not suffer an overfitting problem. This also means the two models can generalize well for new data.

the other metric used to measure each model's performance are training loss training loss used to measure the inconsistency of the predicted value and actual value for all the yolov5 model, vanilla LSTM and Bi-LSTM. The mean training loss for the Yolov5s, vanilla LSTM and Bi-LSTM are 1.3%, 3.2%,13% respectively. Furthermore, the mean validation loss for the proposed models is 1%, 2%, 26%respectively.

After training the yolov5 models, Vanilla LSTM and Bi-LSTM tested with unseen data to compare the model ability to generalize. From the experiment, the test accuracy of the unseen data using yolov5s for image, vanilla LSTM and Bi-LSTM for sensor data is 96.2%, 94%, 95% respectively. Based on the result the proposed models yolov5 can classify nicely and Bi-LSTM better than vanilla LSTM. Since the proposed model pass through several testing finally, the model performs is better when tested with on test set and real-time.

CHAPTER 5 : Conclusion and Recommendation

5.1 Conclusion

There are millions of people in Ethiopia that suffer from hearing loss. These folks require ETHSL as a means of communication among themselves. They do, however, live alongside others who communicate using spoken language. Hearing impaired people lack the ability to use spoken language effectively, whereas hearing people lack the ability to translate the ETHSL. Hearing-impaired people's lives are made much more difficult by this chasm. Local research initiatives to bridge the gap between those folks are really beneficial. This research is a part of an effort to overcome the problem of hearing-impaired persons communicating with hearing people.

The goal of this research is to create and implement a system that can recognize Ethiopian sign language alphabet in real time. The system has two main parts: The vision part which is implemented using yoloV5, and the sensor part which is implemented using LSTM. And by decoupling dynamism of EMA in two separate parts so that we can detect and classify in realtime.

The system begins by accepting video frame and Sensor data from smartwatch as of an Ethiopian alphabet sign. A video was captured with a participant's mobile camera with different size, background and light. The sensor data was captured with TicWatch 2 smartwatch that have a wear app that sends 3 axis Acc and Gyro data to local Serve. Therefore, hand-designed feature extractor is not very robust the yolo model will hand it very efferently, also some tasks, like gray scale conversion, cropping, sharpening and contrast adjustment. Yolo will hand all the object detection part with its deep network. At the same time the LSTM model will sample the sensor data at 50hz and feed to the model.

In this research work, we did different experiment for the sensor data. we have trained two LSTM models. Vanilla LSTM give 96.26% mean accuracy, 89.21% mean validation

accuracy and test accuracy of 95.1. Bi-LSTM give 96.26% mean accuracy, 89.21% mean validation accuracy and test accuracy of 95.1. the yolov5 model that is want to classify the hand shape gives 97.1% mean training accuracy, 94.4 mean validation accuracy and test accuracy 98.5. thus, we decouple the two parts of ema and model the separately with the inference time of yolov5s 1.4ms and 2ms, by running simultaneously the two models are fused by their result.

5.2 Recommendation

In this thesis, we attained a result in recognizing Ethiopian sign language and it has its own impact on the Ethiopian sign language. However, there are gaps which should be filled in the future papers because the work cannot be used as a full translation system of Ethiopia Sign Language.

- It is the researcher's recommendation to design a system that can recognize both EthSL EMA and word to come up with a full-fledged system.
- Our work presented about one-way communication which means it only translates the sign to text. Therefore, we suggested that other researcher will design the system which should work like two-way communication to translate sign to text and vice versa.
- It is the researcher’s recommendation to reduce number of parameters and modify yolov5.
- Researches should be expanded to the implementing this work in mobile device.

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